

COBORDISM MAPS AND ATIYAH–FLOER ISOMORPHISMS FOR FRAMED INSTANTON HOMOLOGY II: INTEGER COEFFICIENTS

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ABSTRACT. We study the extension to integer coefficients of the cobordism comparison between framed instanton homology and the Lagrangian Floer homology of a Heegaard splitting. Orientations are transported from almost-complex orientations on the gauge side through moduli spaces of the mixed anti-self-duality and holomorphic curve equation. Subject to explicit analytic hypotheses on completed mixed domains and oriented gluing, we obtain a projective natural isomorphism for connected cobordisms with a framed arc and a bundle trivialized at the ends. The symplectic maps are signed counts of the triangles, compression cups and caps used in the mod-two comparison. The principal additional arguments concern the constancy of sphere-gluing signs, reflection of coherent orientations, and the identification of the diagonal compression cup with continuation. We keep the grading signs in the gauge orientation convention explicit. The result allows one overall sign per cobordism; comparison with homology-oriented integral instanton maps and with relative-spin quilt orientations is left open.

First draft. The principal comparison is conditional on Hypotheses 2.1, 2.2, and 2.3. Their verification remains part of the integral extension.

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1. INTRODUCTION

1.1. The integral comparison. Let M be a closed connected oriented three-manifold. The framed instanton construction replaces M by $M^\# = M \# T^3$ and uses an $\mathrm{SO}(3)$ bundle which is trivial on M and odd on a torus in the added summand. A Heegaard splitting determines two Lagrangians in the moduli space of flat connections on the stabilized surface. Daemi, Fukaya and Lipyanskiy construct an isomorphism between the resulting gauge and symplectic Floer homologies by counting solutions of a mixed equation [1]. The mixed equation couples the anti-self-duality equation on a four-dimensional region to a holomorphic curve equation on a surface; the two solutions have matching flat boundary values.

Paper I [17] concerns compatibility with four-dimensional cobordisms over $\mathbb{F}_2$. Its symplectic maps count holomorphic triangles and compression cups and caps. The two analytic comparisons there are a collapse of a compression body next to a holomorphic strip and an excision along a torus in a mixed domain. The purpose of this paper is to give the orientation arguments required for integer coefficients. We retain those geometric maps: the symplectic cobordism map is not defined by conjugating an instanton map through an isomorphism.

There are three distinct orientation questions. First, the quotient by the lifting class used in framed instanton homology must preserve orientations. Second, the collapse and torus-excision identifications must preserve the orientations of individual solutions. Third, the orientation system transported through a reflected mixed domain must be compared with the dual of the original system. The last comparison is needed to turn the tensor supplied by the compression cup into a map with the source complex used in Paper I. An unspecified sign on each generator would not suffice: the ambiguity in a cobordism map must be a single sign.

We use the almost-complex orientation construction of Kronheimer–Mrowka [6, Section 5.1]. It is invariant under bundle automorphisms. The almost-complex structure on a cobordism is fixed near the odd torus, but its restrictions at the two ends are allowed to vary. Conversion operators then identify the end orientation lines with fixed object lines. Their placement relative to the generator lines matters when a conversion is odd. We denote the resulting gauge functor by $I_j^\#$ to record this convention.

1.2. Statement and hypotheses. The category $\mathbf{B}$ is described in Section 2. An object is a closed connected oriented three-manifold with the framing data needed for its torus summand. A morphism is a connected oriented cobordism W equipped with a normally framed arc ν joining its ends and an $\mathrm{SO}(3)$ bundle E trivialized at the ends and along the arc. A presentation p consists of a Heegaard splitting, metric data, small primary holonomy perturbations, and regularizing data. The additional smallness and reflection conventions for integer coefficients are specified in Section 13.

Write $C_G(M, p)$ for the integral gauge complex with the almost-complex orientation convention, and $CF(M, p)$ for the symplectic complex with transported orientations. The graded homotopy category $\mathcal{K}(\mathbb{Z})$ and the notation $F \simeq \pm G$ are defined in Section 3. For an admissible decomposition d of a cobordism, $S_d(W)$ is the composite of signed holomorphic triangle maps, compression cups, compression caps, and continuation maps.

The geometric arguments use the unsigned compactness and gluing results of Paper I and the following additional hypotheses. They are stated precisely in Section 2.4.

- (i) Uniform regularity and compatible Fredholm realizations on the completed mixed domains, including families, changes of weights, and the constant reference configurations.

- (ii) An oriented torus-completion comparison, compatible with ordinary gluing and with the reference configurations used to normalize the vertex maps.
- (iii) Determinant-one self-gluing along the nonseparating hypersurface in the gauge excision cobordism, with its gluing multiplicities and determinant-line identifications.

These are mathematical hypotheses, not assertions that follow merely from the existence of the corresponding mod-two count. In particular, the first draft does not claim an unconditional integral version of the mixed excision theorem.

Theorem A (Conditional integral comparison). *Assume the analytic results of Paper I and Hypotheses 2.1, 2.2, and 2.3. With the orientation and admissibility conventions of this paper, the mixed maps*

$$N_p : C_G(M, p) \longrightarrow CF(M, p)$$

are isomorphisms of finite free integral complexes. For every morphism $(W, E, \nu) : M_0 \rightarrow M_1$, admissible endpoint presentations p_0, p_1 , and admissible decomposition d , one has

$$(1.1) \quad N_{p_1} I_J^\#(W, E, \nu) \simeq \pm S_d(W, E, \nu) N_{p_0} \quad \text{in } \mathcal{K}(\mathbb{Z}).$$

The sign is one sign for the entire map. Consequently the symplectic maps are independent of the decomposition up to sign and chain homotopy. On homology they define a projective functor and $H(N)$ is a projective natural isomorphism from $I_J^\#$ to that functor. Each symplectic homology map has the parity of the corresponding gauge map.

At objects with nonzero integral gauge homology the symplectic maps are the geometric maps of Paper I with the orientations constructed here. If the integral gauge homology vanishes, the complex is contractible; maps incident to it represent the zero morphism in $\mathcal{K}(\mathbb{Z})$, with the convention of Section 12.5. Reduction modulo two recovers the maps of Paper I, up to chain homotopy in the contractible case.

The qualifier “projective” has the usual literal meaning: maps are identified with their negatives. No system of signs making all composition identities exact is asserted. In particular, multiplying individual elementary maps by signs is permitted, whereas multiplying different matrix entries by unrelated signs is not.

1.3. The two orientation comparisons. The proof of Theorem A separates two further statements. They use the same hypotheses as that theorem.

Theorem B (Orientations under collapse). *The bijection of regular index-zero solutions in the compression-body collapse of Paper I preserves the transported determinant-line orientations. This assertion uses the metric and the admissible linearizations specified in Section 5. It is compatible with ordinary gauge and strip gluing and with compact parameter families. The two-handle comparison therefore holds over $\mathbb{Z}$, up to the single sign arising from the auxiliary orientation data of the cobordism.*

Theorem C (Reflection and the diagonal cup). *Fix a Heegaard splitting, metric data, and an almost-complex base structure at an object with nonzero integral gauge homology. In the admissible ball of primary perturbations, the sphere character of the transported strip orientation system is constant and agrees with that of the reflected presentation. There is consequently a reflection isomorphism*

$$\Phi_p : CF(M, \bar{p}) \longrightarrow CF(M, p)^\vee$$

determined by the configuration-level orientations up to one sign. With this pairing, the compression cup with no handle attached is chain homotopic, up to one sign, to the symplectic continuation map.

The sphere character records the change of a coherent orientation when a sphere is attached to a strip configuration. Its constancy does not compute its value. The reflection statement compares two characters; it does not assert that either character is trivial. This distinction avoids imposing an unnecessary relative-spin orientation convention on the compression correspondence.

1.4. Strategy. The integral mixed map has a unit diagonal in the filtration used in Paper I. The diagonal configuration is a constant pair, and its orientation is the canonical orientation of an invertible operator. The resulting integral isomorphism is stronger than a mod-two isomorphism: an odd integer coefficient on the diagonal would not be enough.

For two-handles, determinant lines are transported through the collapse family. The relevant continuity is continuity over smooth configurations with all derivatives controlled on compact subsets and with the prescribed asymptotic behavior. Strong convergence of solutions alone does not identify all operator families used in an orientation argument. We state the needed realization and completion properties explicitly and prove the sign comparison from them.

For one-handles, a folded triangle gives a tensor in the product of the reflected source complex and the target complex. Excision identifies this tensor with a gauge handle state. This gives a cup formula before any reflection pairing is inserted. The remaining problem is to identify the pairing with the one for the actual symplectic continuation maps. We first prove that the sphere character is constant under small changes of primary data, then compare it across a fold. A rigidity argument for coherent systems identifies the diagonal cup with continuation. The cup formula now gives small-primary naturality, mixed duality, and the one-handle square. Reflection gives the three-handle square.

Finally, the algebraic descent uses chain homotopies throughout. Over $\mathbb{Z}$, equality on homology is generally insufficient to prove that two chain maps are homotopic. The proof therefore retains the homotopies supplied by the one-dimensional moduli spaces and uses the invertibility of the mixed maps only after the elementary squares have been established.

1.5. Scope of the draft. The almost-complex convention used for $I_J^\#$ is part of the statement. We do not identify it with Scaduto's homology-oriented integral cobordism maps [11]. An odd conversion can introduce a grading operator rather than an overall sign. Section 4 explains this point and the additional grading information needed for a comparison. We likewise do not identify N_p with the integral map of [1]; their reductions modulo two are compared in Paper I at common admissible data.

The orientations on holomorphic curves are transported from the gauge side. Comparison with the relative-spin orientations used in quilted Floer theory [14, 16] is a separate question. The underlying curves and correspondences are the same ones as in Paper I, but an equality of mod-two counts does not supply this integral orientation comparison. Disconnected objects, units, counits, closed pairings, and nontrivial bundles on the objects are outside the present statement. Bundles on cobordisms may be nontrivial relative to the specified trivialized ends.

2. GEOMETRIC DATA AND ANALYTIC HYPOTHESES

2.1. Framed objects and cobordisms. Fix an oriented ball in M and form $M^\# = M \# T^3$ along that ball. Write $T^\circ = T^3 \setminus \text{int } B^3$. Choose a torus $T_0 \subset T^\circ$, a neighborhood $N(T_0)$, and an odd bundle $P_T \rightarrow T^3$ with $\langle w_2(P_T), [T_0] \rangle = 1$. The bundle $P_M \rightarrow M^\#$ is obtained by joining P_T to the trivial bundle on M . The lifting class

$$\alpha_M = \text{PD}[T_0] \pmod{2} \in H^1(M^\#; \mathbb{Z}/2)$$

acts on the determinant-one configuration space; the framed complex uses the quotient by $\phi_M = \langle \alpha_M \rangle$.

For (W, E, ν) as in the introduction, remove a tube around the normally framed arc and insert $T^\circ \times [0, 1]$. Denote the result by $(W^\#, E^\#)$. The lifting subgroup ϕ_W is the one induced by this torus tube, with the endpoint restrictions ϕ_{M_i} . At an intermediate disconnected cut we retain the full subgroup specified by the tube; it need not be the product of the groups on the components. This distinction is essential in the excision calculation in Section 8.

The category identifies isomorphic decorated cobordisms and uses gluing along the fixed endpoint data for composition. The connected source and target condition permits a handle decomposition with one-, two-, and three-handles, together with changes of presentation. As in Paper I, elementary words also include metric changes, Heegaard moves and frame changes. The presentation graph is connected and the gauge cobordism map is independent of such a word. These are the geometric inputs to the descent at the end of the paper.

2.2. Compression bodies and the symplectic complex. If $M = H_0 \cup_\Sigma H_1$ is a Heegaard splitting of genus g , adjoin $[0, 1] \times T^2$ to each handlebody. The resulting torus compression bodies C_i have boundary $\Sigma_h \sqcup T^2$, where $h = g + 1$. Let $\mathbf{M}_h$ be the moduli space of flat connections on the odd $\mathrm{SO}(3)$ bundle over Σ_h , with the lift convention of [1, Section 2]. Its symplectic form is the usual Atiyah–Bott form. Restriction of flat connections on C_i gives a Lagrangian $L(C_i) \subset \mathbf{M}_h$.

Primary perturbations h_i are finite sums of averaged holonomy functions supported in C_i° , away from the fixed collars. Their critical equations are

$$(2.1) \quad *_3 F_B + \nabla_B h_i = 0.$$

We use a sign-sensitive trace convention for fixed-determinant $\mathrm{U}(2)$ holonomy: on each solid torus choose a continuous square root of the determinant holonomy and take the trace of the resulting $\mathrm{SU}(2)$ part. The choices are made once for the finite family. Equation (2.1) determines perturbed Lagrangians $P_i = L_{h_i}(C_i)$. At transverse data, their intersection points correspond to the gauge critical orbits.

For each intersection point x , let a_x be the corresponding gauge critical point and $L_G(a_x)$ its real determinant line. For a real line L , write $\mathfrak{o}(L)$ for the rank-one free abelian group generated by its two orientations with the relation $[-o] = -[o]$. The integral symplectic chain group is

$$(2.2) \quad CF(M, p) = \bigoplus_{x \in P_0 \cap P_1} \mathfrak{o}(L_G(a_x)).$$

The differential counts rigid strips with the transported orientation system of Section 6. The mod-two reduction of (2.2) is the chain group of Paper I.

2.3. Mixed configurations. A mixed domain consists of a gauge region and a symplectic surface meeting along a matching boundary. On the gauge side a connection satisfies a perturbed anti-self-duality equation. On the symplectic side a map to $\mathbf{M}_h$, or to a product of such spaces, satisfies a perturbed Cauchy–Riemann equation. Along a matching boundary the surface restriction of the connection is flat and represents the value of the map. At ordinary ends we prescribe irreducible nondegenerate gauge limits or transverse Lagrangian intersection points. At mixed ends we retain the finite-dimensional stationary coordinates explicitly.

The domains used below are the mixed vertex domain, mixed and folded triangles, their strip gluings, the closed-cut mixed domain, the finite-width torus domains, their completed

limits, and the collapse families. We also use ordinary strips, continuation strips, and the disks obtained by doubling and capping a folded triangle. Statements about determinant lines apply on configuration spaces, not just on their solution subsets.

We use the smooth topology on these spaces: in fixed end trivializations all derivatives converge on compact sets, the asymptotic limits converge with their parameters, and the decaying parts converge in the chosen weighted norms at every derivative order. A continuous compact family means a family continuous in this topology. Sobolev completions are used locally to construct Fredholm operators and determinant lines. They do not replace the smooth topology when a holonomy family is differentiated repeatedly.

2.4. The analytic assumptions. The unsigned moduli spaces, their energy-index estimates, and the low-dimensional compactifications are those of Paper I. The following assumptions specify the additional uniform and oriented statements used in this draft. The ordinary local Fredholm and regularity results for the mixed equation are those of [1, 2]; the hypotheses below include their use on the additional completed domains of this paper.

Hypothesis 2.1 (Uniform realizations and regularity). For each mixed domain listed above, and for each compact parameter family used below, the following hold.

- (a) Smooth configurations with fixed asymptotics admit local matching-boundary charts. The gauge-fixed linearization on weighted spaces is Fredholm; its kernels and cokernels are smooth. Changes of chart and the admissible changes of realization of Definition 5.1 give continuous families of Fredholm operators and the same determinant line.
- (b) At ordinary ends the weights lie in common spectral gaps. At mixed ends the stationary coordinates are separated from the decaying part. Regular zero-dimensional solutions and the constant reference pairs have invertible realized operators. On compact families the estimates, slice radii, and smallness thresholds can be chosen uniformly, including at infinite torus length.
- (c) The mixed shifting construction, its relative family form, and node gluing are continuous in the smooth topology. They commute with ordinary end gluing. A compact family of shifted configurations can be made exactly symplectically constant outside one common compact set, without changing a member already constant there.
- (d) The solution and homotopy counts are independent of the choice of weights within the specified gaps. The pure symplectic continuation, cap, and strip-gluing moduli spaces have the usual regular low-dimensional compactifications, with no additional bubbling boundary in the dimensions used here.

The relevant asymptotic restriction is strict: the weight is less than the smallest positive decay rate of every end occurring in that instance. Uniformity is asserted for a fixed compact family, not over all Heegaard splittings or all primary perturbations. When an end is completed, the same statement must hold on the completed domain. A finite-neck estimate by itself does not imply this last assertion.

Hypothesis 2.2 (Oriented mixed torus completion). Consider the torus domains in the proof of the mixed excision theorem of Paper I, with fixed admissible primary and secondary data. The determinant lines extend continuously across the width and completion comparisons. Their identifications have the following properties:

- (a) they are natural under the lifting groups and compatible with gauge and strip gluing, including the conversion lines at the closed cut;

- (b) at a split constant reference pair they are the product of the two canonical invertible-operator orientations;
- (c) the preglued configurations and the solutions are connected by paths in the stated charts which identify these determinant lines, with the same property for the one-dimensional families used to compare lengths and data;
- (d) the compactified one-dimensional comparison has only the broken ends and endpoint strata used in Paper I. With outward normal first and the parameter orientation fixed, its boundary gives the integral chain homotopy asserted in Theorem 8.4.

This assumption concerns an oriented comparison of a particular family of mixed equations. It does not assume the cup formula, reflection invariance, or the cobordism comparison. It is precisely the extension of the torus-completion argument for which a complete integral verification is still needed.

Hypothesis 2.3 (Determinant-one self-gluing). Let X be a gauge cobordism occurring in the torus-excision construction, and cut it along its designated closed hypersurface Y . Suppose the limiting critical points on Y are irreducible and nondegenerate and the data are translation invariant on the neck. For sufficiently large neck length the index-zero solutions on X are obtained by gluing the matched determinant-one solutions on the cut cobordism. The fiber of this gluing has the lifting multiplicity described in Lemma 8.2. There are no additional index-zero limits. The determinant-line gluing uses the graded symmetry of Section 3; its sign depends only on the ordered end degrees and the fixed orientation choices. The corresponding one-dimensional families give the composition homotopies.

Ordinary composition and trace constructions in instanton Floer theory provide the models for this hypothesis [6, Sections 5.2–5.3], [7, Section 2.5]. Here its nonseparating form and the precise determinant-one multiplicity are part of the assumption. We do not infer this form from a trace in a quotient complex.

3. COMPLEXES, GRADED SIGNS AND DETERMINANT LINES

3.1. The homotopy category. A complex is a finite free $\mathbb{Z}/2$ -graded abelian group $C = C^0 \oplus C^1$ with an odd differential d_C . For a homogeneous map $F : C \rightarrow D$ of degree f , set

$$(3.1) \quad \delta F = d_D F - (-1)^f F d_C.$$

A closed map satisfies $\delta F = 0$. Thus an odd closed map anticommutes with the differentials. We allow sums of the two degrees when discussing homotopies of the composites below. The category $\mathcal{K}(\mathbb{Z})$ has these complexes as objects and closed maps modulo boundaries as morphisms. In particular

$$F \simeq G \iff F - G = \delta K.$$

For homogeneous F, G of degree f , the homotopy K can be chosen of degree $f + 1$. The notation $F \simeq \pm G$ means that one of $F - G, F + G$ is a boundary, with a single choice of sign.

If A and B are closed and B has degree b , then

$$(3.2) \quad (\delta K)A = \delta(KA), \quad B\delta K = (-1)^b \delta(BK).$$

These identities justify composition of every homotopy used below. They also show that projectivizing by the relation $[F] = [-F]$ is compatible with composition.

Lemma 3.1. *An acyclic finite free $\mathbb{Z}/2$ -graded complex over $\mathbb{Z}$ is contractible. An even quasi-isomorphism between finite free complexes is a chain homotopy equivalence.*

Proof. Let $B^i = \text{im}(d : C^{i-1} \rightarrow C^i)$. Each B^i is free as a subgroup of a finitely generated free group. If the complex is acyclic, the exact sequence $0 \rightarrow B^i \rightarrow C^i \xrightarrow{d} B^{i+1} \rightarrow 0$ splits. Relative to these splittings the differential identifies the complementary summand of C^i with B^{i+1} . Its inverse on this summand, and zero on the complement, defines h with $dh + hd = 1$. Apply this to the mapping cone of a quasi-isomorphism to obtain the second assertion. $\square$

The lemma does not say that a map inducing zero on homology is null-homotopic. Nor does it identify arbitrary free complexes with their homology, which may contain torsion. We use the lemma only for acyclic complexes and actual quasi-isomorphisms.

3.2. The tensor convention. Generator lines precede the determinant lines of trajectories and cobordisms. With this ordering the tensor differential is

$$(3.3) \quad d(x \otimes y) = (-1)^{|y|} dx \otimes y + x \otimes dy.$$

For homogeneous maps F, G we use

$$(3.4) \quad (F \otimes_R G)(x \otimes y) = (-1)^{|F||y|} Fx \otimes Gy.$$

Closed maps tensor to a closed map of degree $|F| + |G|$ with this rule. The usual left tensor differential is related to (3.3) by the even chain isomorphism

$$T(x \otimes y) = (-1)^{|x||y|} x \otimes y.$$

Thus a formula imported in a left tensor convention is conjugated by T at each tensor end before it is used here.

Lemma 3.2. *For a disconnected gauge end $Y_1 \sqcup Y_2$, with the generator line ordered as $L(a_1) \otimes L(a_2)$, gluing index-one trajectories on the two components gives (3.3).*

Proof. A trajectory on Y_1 has an odd determinant line. To place it next to $L(a_1)$, move it past $L(a_2)$; this gives $(-1)^{|a_2|}$. A trajectory on Y_2 is already adjacent to its generator line. Summing the two contributions gives the formula. $\square$

Put $\varepsilon_C x = (-1)^{|x|} x$. If a homogeneous map F of degree f commutes strictly with the differentials, the map $F^r = F \varepsilon_C^f$ is closed. For strictly commuting maps F, G of degrees f, g , respectively,

$$(3.5) \quad (FG)^r = (-1)^{fg} F^r G^r.$$

The discrepancy is one sign per composite. By contrast the factor ε_C is not generally an overall sign.

For example, on $C^0 = C^1 = \mathbb{Z}$ with zero differential, the map interchanging the two generators and its product with ε_C are not equal up to sign. The difference persists on homology. This elementary example explains why an integral comparison must specify the convention for odd cobordism maps.

3.3. Duals, evaluation and tensors of maps. The graded dual has

$$d^\vee f = -(-1)^{|f|} f \circ d, \quad F^\vee(f) = (-1)^{|F||f|} f \circ F.$$

The canonical even double-dual map is

$$\text{ev}_C : C \longrightarrow C^{\vee\vee}, \quad \text{ev}_C(x)(f) = (-1)^{|x||f|} f(x).$$

The sign is necessary: the unsigned double-dual identification does not commute with the differential in this convention. There is an unsigned chain isomorphism

$$(3.6) \quad \beta : C^\vee \otimes_R D \longrightarrow \text{Hom}(C, D), \quad \beta(f \otimes y)(x) = f(x)y.$$

Indeed, substituting the dual differential in (3.3) gives exactly (3.1) after evaluation on a homogeneous x . For a tensor in $C \otimes_R D$ we write

$$B_C = \beta \circ (\text{ev}_C \otimes_R 1) : C \otimes_R D \longrightarrow \text{Hom}(C^\vee, D).$$

If e_i is a homogeneous basis of C , the tensor corresponding to a homogeneous map $F : C \rightarrow D$ is

$$\text{ten}(F) = \sum_i e_i^\vee \otimes F(e_i).$$

This expression is independent of the chosen basis. It is a cycle if and only if F is closed. For $F = 1$, it is the coevaluation tensor in the present convention.

3.4. Determinant lines. For a real Fredholm operator $D : X \rightarrow Y$ set

$$\det D = \Lambda^{\max} \ker D \otimes (\Lambda^{\max} \text{coker } D)^*, \quad |\det D| = \text{ind } D \pmod{2}.$$

We use the graded determinant convention of [18]. Stabilization by a finite-dimensional map $V \rightarrow Y$ with $D \oplus V$ surjective gives the local trivializations. Enlarging the stabilization gives compatible transition maps. This construction applies to continuous families of Fredholm operators without requiring constant kernel dimension.

For direct sums the determinant lines are ordered in the order of the summands. Interchanging two summands gives $(-1)^{\text{ind } D_1 \text{ind } D_2}$. The dual of a tensor reverses the order of its factors and uses nested evaluation. An invertible operator has the canonical determinant orientation of $\mathbb{R}$; a continuous path of invertible operators preserves this orientation.

Lemma 3.3 (Ordinary gluing). *Suppose two Fredholm problems have matching ordinary ends with invertible asymptotic operator. Gluing with a sufficiently long neck gives a determinant-line isomorphism*

$$\det D_1 \otimes \det D_2 \longrightarrow \det(D_1 \# D_2)$$

well defined up to a positive scalar. It is continuous in compact parameter families, natural under bundle automorphisms, and associative for disjoint necks. On invertible operators it is the canonical identification.

Proof. Choose finite-dimensional stabilizations supported away from the ends so that the two stabilized operators are surjective. Patch their right inverses using cutoffs. The error tends to zero with the neck length, so a Neumann series corrects the patched operator to a right inverse. Projection of the patched kernels to the kernel of the glued stabilized operator gives an isomorphism for long necks. Its top exterior power and the stabilization isomorphisms define the asserted map.

Enlarging stabilizations commutes with this construction. Changing cutoffs, projections or neck length gives a path of kernel isomorphisms, so the sign does not change. For two disjoint necks use stabilizations supported away from both; the two successive patching maps are homotopic to the direct three-piece patching map. This proves associativity, with only the graded signs caused by an actual reordering. When both operators are invertible, no stabilization is needed and the map is $\mathbb{R} \otimes \mathbb{R} \rightarrow \mathbb{R}$. The gauge version is the determinant-line construction accompanying ordinary Floer gluing [4, Proposition 5.11]; the strip version uses the same stabilization argument for real Cauchy–Riemann operators [5]. $\square$

Gluing a constant cylinder is the positive identity: its operator is invertible, and the patched kernel map converges to the identity as the neck grows. Weighted gluing is treated by conjugating with the exponential weight. The resulting asymptotic operator must be

invertible, and matched ends must use compatible weights. We will never use ordinary gluing at a mixed end without separately keeping its stationary coordinates.

4. THE INTEGRAL GAUGE CONVENTION

4.1. Almost-complex orientations with free ends. Choose an S^1 -invariant almost-complex germ J_T near $S^1 \times N(T_0)$, with T_0 an almost-complex torus in its specified orientation. On an object, such a structure is equivalently specified by a nonvanishing vector field on $M^\#$, normalized near T_0 . Choose a base J_M^0 for each object. On a cobordism require J_W to agree with the fixed germ near $N(T_0) \times [0, 1]$ and to be translation invariant on its end collars. Its end restrictions J'_0, J'_1 are not prescribed in advance.

Lemma 4.1. *Every decorated cobordism admits an almost-complex structure of this kind.*

Proof. The fiber of the bundle of compatible oriented almost-complex structures is $\mathrm{SO}(4)/\mathrm{U}(2) \cong S^2$. Let $A = N(T_0) \times [0, 1]$. The primary relative obstruction lies in $H^3(W^\#, A; \mathbb{Z})$ and maps to $W_3(W^\#) = 0$. Here an oriented four-manifold admits a spin^c structure; equivalently its second Stiefel–Whitney class has an integral lift. The remaining relative class is in the image of the connecting map from $H^2(A; \mathbb{Z})$. The restriction $H^2(W^\#; \mathbb{Z}) \rightarrow H^2(A; \mathbb{Z})$ is surjective: the torus summand supplies a class pairing once with T_0 , and this class extends by the Mayer–Vietoris sequence across the three-sphere where the tube is attached. Thus the connecting map vanishes.

The secondary obstruction lies in $H^4(W^\#, A; \mathbb{Z})$. Since $A \simeq T^2$, its third cohomology vanishes; since $W^\#$ is connected with nonempty boundary, $H^4(W^\#; \mathbb{Z}) = 0$. The relative group therefore vanishes. Homotopy extension makes the structure equal to the prescribed germ, and a collar reparametrization makes it translation invariant near the ends. $\square$

Prescribing both end structures would change this obstruction problem. We use Lemma 4.1 only with free end restrictions. Conversion maps, rather than an extension assertion with fixed ends, implement the fixed object convention.

The almost-complex deformation of the anti-self-duality operator has a complex-linear endpoint. The complex orientation of its kernel and cokernel orients its determinant line. This construction is invariant under the full group of bundle automorphisms, which act complex linearly at that endpoint [6, Section 5.1]. Ordinary determinant gluing preserves these orientations. In particular the action of ϕ_M on the orientation lines has no additional sign twist.

4.2. The comparison character. Let v, v' be the normalized vector fields corresponding to J, J' on $S^1 \times Y$, where $Y = M^\#$. Their difference class $\delta(v, v') \in H^2(Y; \mathbb{Z})$ satisfies

$$2\delta(v, v') = c_1(\xi_{v'}) - c_1(\xi_v).$$

The difference class is defined by the obstruction to homotoping the vector fields on the two-skeleton, rather than by choosing an arbitrary half of the displayed Chern-class difference. This is relevant when $H^2(Y; \mathbb{Z})$ has two-torsion.

Proposition 4.2. *If $\alpha \in H^1(Y; \mathbb{Z}/2)$ is the reduction of an integral class, the change of almost-complex orientation around a loop with lifting class α is*

$$(4.1) \quad \xi_{J, J'}(\alpha) = (-1)^{\langle \delta(v, v'), \mathrm{PD} \alpha \rangle}.$$

In particular it is 1 on ϕ_M when J and J' agree near T_0 .

Proof. Represent the loop by a path of connections whose endpoints are identified by a bundle automorphism of class α . Closing up gives a bundle P_α on $S^1 \times Y$ with

$$w_2(P_\alpha) = \text{pr}_Y^* w_2(P) + [dt] \smile \alpha.$$

The determinant-line comparison for this closed operator is divided by the comparison for the constant loop. The P -independent orientation factor therefore cancels.

For completeness, the bundle-dependent comparison can be computed on a reducible model $\mathbb{R} \oplus L$. The anti-self-dual operator on the L -summand has both the complex structure induced by J and that of L . On the summand where these structures are opposite, conjugation changes the determinant orientation by $(-1)^{\text{ind}_\mathbb{C}}$. The index formula gives, for two almost-complex structures, the ratio

$$(-1)^{\langle c_1(L) \smile \delta(v, v'), [S^1 \times Y] \rangle}.$$

This is the relative form of the usual almost-complex orientation calculation[3, Section 3(c)]. Changing the instanton number does not change the ratio: instanton addition preserves both complex orientations.

Because α has an integral lift, so does the displayed Stiefel–Whitney class. The reducible model therefore computes the desired ratio. Only the term $[dt] \smile \alpha$ survives after division by the constant-loop comparison; integration over the circle gives (4.1). The closing-up identification is made in pair form $\det D \otimes (\det D_{\text{const}})^*$, so it introduces no choice of orientation at the closed end. It is the real excision identification, independent of J and J' . Finally α_M is dual to T_0 , and the difference class restricts to zero there when the structures have the same germ. $\square$

Only integral lifting classes are used in this proposition. The lifting classes in the framed objects have that property. No assertion for arbitrary mod-two classes is needed.

4.3. Conversion maps and their ordering. For a critical point a , let Q_a interpolate the complex operators for J and J' . Choose a weight for which this operator is Fredholm and choose an orientation q_a of its determinant line. Transport q_a along paths of connections. Proposition 4.2 says that this transport is consistent on the framed quotient. It is unique up to one sign after a choice at one reference connection.

We place q_a after the generator line and obtain

$$(4.2) \quad \Theta_{J \rightarrow J'}^r : C_G(Y, J) \longrightarrow C_G(Y, J'), \quad x \longmapsto \gamma(x \otimes q_a).$$

Its parity $t = \text{ind } Q_a \pmod{2}$ is independent of a . Moving the odd trajectory line past q_a shows that

$$d\Theta^r = (-1)^t \Theta^r d.$$

Thus it is a closed isomorphism of degree t . If Θ^1 denotes the convention with q_a placed before the generator, then

$$(4.3) \quad \Theta^r = \Theta^1 \varepsilon^t.$$

Composition of conversion maps agrees with the direct conversion up to one sign. The sign is constant because the orientation comparison is defined over the connected space of connections, not chosen separately at its critical points.

Definition 4.3. For (W, E, ν) choose an admissible J_W with end restrictions J'_0, J'_1 . Define

$$(4.4) \quad I_J^\#(W) = \Theta_{J'_1 \rightarrow J_{M_1}^0}^r I(W^\#, E^\#, J_W)^{\phi_W} \Theta_{J_{M_0}^0 \rightarrow J'_0}^r.$$

At the induced end structures the middle map uses the almost-complex orientations and the stated ordering of its determinant line. Tensor ends use (3.4).

Proposition 4.4. *The homotopy class of (4.4) is independent of J_W up to one sign. These classes define a projective gauge functor, and their reduction modulo two is the framed gauge functor of Paper I.*

Proof. Compare two choices of J_W after inserting their endpoint conversions. They orient the same real determinant line over the space of connections with fixed endpoint limits. Their ratio is continuous and takes values in $\{1, -1\}$, so it is constant on that space. Gluing a path at either endpoint shows that the ratio does not change when the limiting critical point is changed. The character calculation makes this assertion invariant under changing the lift of an end critical point. Thus the ratio is one sign for all entries of the map.

For composition choose an admissible almost-complex structure on the composite and make it cylindrical at the intermediate cut. The restrictions then have identical structures there, and the ordinary gauge composition theorem applies [6, Proposition 5.4]. Insert the conversions to the chosen object bases. The conversions at the common end cancel up to sign, and (3.5) accounts for their ordering. Independence of J_W gives the result for the original choices. The product cobordism gives the identity class. All orientation signs disappear modulo two. $\square$

4.4. Reflection and the relation to other conventions. Choose the base on the oppositely oriented object in pairs:

$$J_{-M}^0 = \Psi_* \rho_* J_M^0.$$

Here ρ reverses the cylindrical direction and Ψ is the fixed identification of the reflected torus-summed presentation with $(-M)^\#$. Take $T_0 = \{\theta_0\} \times T^2$ and the product germ there; require Ψ to preserve the odd torus, its germ, and its lifting class. Gauge reflection then gives an even isomorphism

$$(4.5) \quad \Phi_{G,M} : C_G(-M) \longrightarrow C_G(M)^\vee, \quad \Phi_{G,-M} = \Phi_{G,M}^\vee \text{ ev}.$$

The transpose formula for a reversed cobordism follows by bending its cylindrical ends and applying the same ordered determinant gluing. All constant end lines are even; any remaining change is one orientation sign of the cobordism. This also gives the transpose formula for gauge continuation.

If the conversion lines are instead placed on the left, put $I_1(W) = \Theta_1^1 I(W) \Theta_0^1$. If the two conversions have parities t_0, t_1 , then

$$(4.6) \quad I_J^\#(W) = (-1)^{t_0 t_1} I_1(W) \varepsilon^{t_0 + t_1}.$$

This is an identity of maps in the stated conventions, rather than an assertion that the two maps differ only by a sign. The degree-dependent factor matters on an object whose homology has both parities.

A $\mathbb{Z}/4$ grading compatible with all the orientation data would provide a further regrading $\eta(x) = (-1)^{g(x)(g(x)-1)/2} x$. The elementary identity

$$\eta(g+d)\eta(g)^{-1} = (-1)^{dg+d(d-1)/2}$$

explains how an odd grading operator can then be converted into a conjugation and one sign per arrow. Identifying such a lift with Scaduto's data is not part of this draft. We therefore keep $I_J^\#$ in the theorem and make no integral identification on the basis of the mod-two equality alone.

5. DETERMINANT LINES ON MIXED CONFIGURATION SPACES

5.1. Admissible operator realizations. Let c be a smooth configuration on a mixed domain, with its specified limits. Denote the mixed equation by $\mathcal{F}(c) = 0$. A chart Φ centered at c incorporates the exact matching condition and the normal seam condition. Its differential identifies a fixed linear boundary problem with the tangent space to the configuration space.

Definition 5.1. An admissible realization of the linearized problem at c is an operator of the form

$$(5.1) \quad L_c = (\mathcal{S} d\mathcal{F}_{\Phi(0)} d\Phi_0, \lambda^2 d_{A'}^{*,g'}).$$

It acts on the prescribed weighted spaces with stationary mixed-end coordinates included. Here $\mathcal{S}$ is a continuously chosen invertible target identification, equal to the identity outside a compact set; A' is a smooth slice center differing from the connection in c only on a compact core and sufficiently little in C^0 ; $g' = f^2 g$ with $f = 1$ off the collapse region and f depending only on base coordinates near a seam; and $\lambda > 0$ equals one outside a compact set. The chart has the same end and seam conventions as the original mixed problem. Changes of all these choices are taken inside their fixed components.

The restriction on the target identification prevents an arbitrary reversal of orientation from being included as a change of realization. The reference identification is the one induced by the geometric charts. Its admissible component is fixed before transporting orientations.

Lemma 5.2 (Realization comparison). *Under Hypothesis 2.1, admissible realizations at a fixed configuration have canonically identified orientation double covers. The identification is continuous in compact families and commutes with ordinary gluing. At a regular zero it preserves the canonical determinant orientation.*

Proof. Changes of charts and target identifications are paths of bounded invertible changes of source and target. They induce the standard naturality maps on determinant lines. Varying a weight inside a gap gives a path of Fredholm realizations; kernel and cokernel elements have the stronger decay common to the two weights, so their identifications are the natural ones. For a change of slice center at a solution, split the tangent space into infinitesimal gauge directions and a fixed complement. Gauge invariance gives $d\mathcal{F}_c d_{A_c} = 0$. The gauge-fixed operator is then block triangular. Its gauge block is invertible by the Poincaré estimate and the smallness of $A' - A_c$; its other block is the intrinsic linearized equation on the quotient. Thus invertibility at a regular zero is unchanged.

Conformal changes in the specified unperturbed region do not change the anti-self-duality equation in four dimensions. They do not change the surface Cauchy–Riemann equation or the normal seam subspace. The same block argument applies to the gauge row. Positive row multiplication does not alter an orientation. These paths give the comparison. A homotopy between two such paths gives an orientation cover over a square, so its endpoint identification is unchanged. The gluing statement follows by choosing the changes away from the ordinary gluing collars and then using continuity. At a regular zero every operator in these paths is invertible, so the canonical orientation is preserved. $\square$

The lemma uses the uniform analytic assertions in Hypothesis 2.1; it is not a proof of their validity on a completed domain. In particular, moving a Coulomb center at a configuration which is not a solution is not justified by the block-triangular argument above. We will use a two-parameter determinant-line comparison when such a configuration occurs in the collapse argument.

5.2. Shifting and relative orientations. A constant pair C_x joins the gauge critical point a_x to the symplectic intersection point x . Its operator is invertible. Let $L_S(x) = L_G(a_x)$. For a mixed configuration c with gauge input a and symplectic output x , shifting and gluing identify it, through a path of configurations, with a configuration of the form $A\#C_x$. Here A is a gauge path from a to an appropriate lift of a_x .

Proposition 5.3. *Under Hypothesis 2.1, there is a relative orientation isomorphism*

$$(5.2) \quad \Gamma_c : L_G(a) \otimes \det D_c \longrightarrow L_S(x),$$

defined up to positive scalar, which agrees with gauge orientation gluing at $A\#C_x$. It is independent of the shifting path, natural under the lifting group, and coherent under ordinary gauge and strip gluing.

Proof. First consider a loop of configurations with fixed limits. The relative family shifting in Hypothesis 2.1(c) deforms it to a loop of symplectically constant configurations. A common gluing length works for the compact family. At that end of the deformation, the determinant bundle is the gauge orientation bundle tensored with the determinant of the invertible constant pair. The gauge bundle is oriented by the almost-complex construction, and the second factor has its canonical orientation. The orientation monodromy of the loop is therefore trivial. This proves path independence of the orientation transported from the constant reference.

For a strip attached at the output, first shift the original mixed configuration to $A\#C_x$. Then shift C_x together with the attached strip to another gauge path followed by C_y . Associativity of determinant gluing identifies both orders of this construction with gauge gluing at the final reference. Continuity transports that equality back to the original configuration. The gauge-end statement is the same argument with the extra path attached before shifting.

The lifting automorphism preserves gauge orientations, conjugates the constant-pair operator, and commutes with gluing. Conjugation preserves the canonical orientation of an invertible operator. Hence the construction descends to the framed quotient. If shifting reaches the other lift of a_x , the two end lines are identified by this same action. $\square$

The pure strip orientation is defined by applying (5.2) to $C_x\#u$ and cancelling the constant pair. This is the transported orientation construction of [1, Section 4.7]. For a mixed triangle there are several gauge inputs; their lines are listed in cyclic order and the same construction is performed on all arms. Associativity is applied before any factor is reordered.

5.3. Parameters and boundary orientations. A parameterized family is oriented by placing the oriented parameter space before the determinant line of the vertical operator. At a regular point this induces the usual orientation of the finite-dimensional zero set. To see continuity when the vertical kernel dimension changes, choose one stabilization on a neighborhood in the parameter space. The orientation is then the exterior-power orientation of the stabilized kernel with the fixed stabilization factor cancelled. Transition maps between such neighborhoods are the determinant-bundle transition maps; no choice of a basis of the moving kernel is required.

For a one-parameter regular family, the boundary orientation is outward normal first. At a strip break the translation vector is the positive vector $\partial_s u$. Moving its odd line past the preceding generator factors gives exactly the signs in (3.3). At a parameter endpoint the two outward normals are opposite. It follows that a compactified one-dimensional family yields an identity of the form

$$(5.3) \quad F_1 - F_0 = \delta K,$$

with the Koszul signs already included in δ . If an orientation for the whole family has been chosen only up to sign, the corresponding endpoint identity has one overall sign. It never has independently chosen signs for its separate matrix entries.

The compatibility of a geometric boundary collar with this determinant-line orientation is part of the gluing assertions in the analytic hypotheses. Merely orienting the interior of a moduli space would not prove (5.3).

6. THE MIXED ISOMORPHISM OVER THE INTEGERS

6.1. Differentials and the vertex map. For an index-one gauge trajectory or holomorphic strip, its sign is positive when the relative orientation agrees with its translation vector. For a regular index-zero mixed solution, its sign is the comparison of (5.2) with the canonical element of its invertible determinant line. Set

$$\begin{aligned}
 d_G o_a &= \sum_b \sum_{[A] \in \mathcal{M}_G^0(a,b)} \text{sign}(A) o_b, \\
 d_S o_x &= \sum_y \sum_{[u] \in \mathcal{M}_S^0(x,y)} \text{sign}(u) o_y, \\
 N_p o_a &= \sum_x \sum_{[c] \in \mathcal{M}_N^0(a,x)} \text{sign}(c) o_x.
 \end{aligned}
 \tag{6.1}$$

The superscript 0 denotes dimension after quotienting translation when that action is present. All sums are finite by the compactness and energy-index bounds of Paper I.

Proposition 6.1. *The transported orientations give $d_S^2 = 0$ and $d_S N_p = N_p d_G$ over $\mathbb{Z}$.*

Proof. The compactified one-dimensional strip space has two-level broken strips as its boundary. Coherence in Proposition 5.3 identifies their boundary orientation with the product of the two zero-dimensional signs, with the sign from the translation factor. The oriented boundary count is zero, yielding $d_S^2 = 0$. For the mixed vertex, the two types of boundary are a gauge trajectory followed by a mixed solution and a mixed solution followed by a strip. The outward-normal orientations of the two ends give opposite signs. Their sum is $d_S N_p - N_p d_G$. The index and energy restrictions exclude additional boundary strata in this dimension. $\square$

6.2. The unit diagonal. The filtration used in Paper I orders the finite generator set so that every index-zero mixed solution goes from a generator to a generator of no larger filtration. At equal filtration the only solutions are the unique small deformations of the constant pairs. This is a statement about the set of solutions, before orienting it.

Lemma 6.2. *Let F be an integral endomorphism of a finite filtered free module. Suppose its associated graded map is diagonal with each diagonal entry in $\{1, -1\}$. Then F is an integral isomorphism. Its inverse preserves the filtration.*

Proof. Write $F = D(1+U)$, with D the diagonal sign matrix and U strictly filtration-lowering. If there are r filtration steps then $U^r = 0$, and

$$F^{-1} = (1 - U + U^2 - \cdots + (-1)^{r-1} U^{r-1}) D^{-1}.$$

All coefficients are integral. $\square$

Each diagonal solution contributes 1 or -1 , since it is one regular point. Thus the invertibility of N_p already follows from Lemma 6.2; it does not require first proving that every

diagonal entry is positive. With the normalization by constant pairs, the entries are in fact +1.

Proposition 6.3. *The map N_p is a chain isomorphism. Its diagonal entries are +1 in the constant-pair normalization. Its homotopy class is independent of regularizing data, after the corresponding gauge continuation is inserted.*

Proof. Take the segment from zero regularizing data to the chosen small data. At a constant pair the linearization is invertible. The chart implicit-function theorem therefore supplies a nearby branch of regular solutions. The uniqueness statement in the triangularity argument identifies this branch with the diagonal solution. The fixed-center regularity in Hypothesis 2.1 makes the branch continuous for determinant-line transport. Its sign is locally constant and begins at the canonical sign +1.

The filtration now gives an inverse integral matrix. Since N_p commutes with the differentials, its inverse does too. Finally connect two choices of regularizing data by a generic small path, allowing the gauge end perturbation to vary and retaining the fixed primary data. The compactified one-dimensional mixed spaces give (5.3). The gauge continuation accounts for the changing gauge complex, while the strip end is unchanged. This proves the stated independence. $\square$

The secondary perturbations are supported away from the strip end, so their variation does not change the strip operator. The segment is chosen in a star-shaped smallness neighborhood where all the reference constant-pair operators remain invertible. This is why an arbitrary path of perturbations cannot be substituted in the normalization argument.

6.3. Parametric orientations. The same reasoning applies to compact parameter families of primary and geometric data whenever their asymptotic intersections remain transverse. Locally choose fixed-center charts and stabilizations. A nearby regular solution has small coordinates in such a chart. The line orientation pulled back from Proposition 5.3 varies continuously, while the canonical orientation of an invertible operator is continuous on the invertible locus. Their ratio is consequently locally constant. This fact will be used in the collapse and closed-cut comparisons. It does not require identifying the topology of every Sobolev quotient with the smooth quotient globally.

7. TRIANGLES AND ORIENTATIONS UNDER COLLAPSE

7.1. The triangle product. Order three Lagrangians L_0, L_1, L_2 by the boundary orientation of the triangle. Write $C_{ij} = CF(L_i, L_j)$. Transport orientations through the mixed triangle and put all generator lines before its determinant line. The resulting product

$$\mu : C_{01} \otimes_R C_{12} \longrightarrow C_{02}$$

satisfies

$$(7.1) \quad d\mu(a, b) = (-1)^{|b|} \mu(da, b) + \mu(a, db).$$

Indeed the first strip break requires its translation line to pass the second input generator; the second break does not. This is the boundary calculation for the one-dimensional triangle space.

If $\theta \in C_{12}$ is a homogeneous cycle, then $R_\theta(a) = \mu(a, \theta)$ is a closed map of degree $|\theta|$. The product itself is even in this convention. For a change of gauge base at any of the three vertices, the corresponding symplectic conversion is diagonal on the same generator lines.

Applying the conversions with (3.4) gives the product in the new bases; no unsigned tensor of two possibly odd conversions is used.

7.2. The determinant comparison at the collapsed limit. Let V be a regular index-zero limiting mixed solution in the collapse theorem of Paper I. Denote its approximate solution on the domain of collapse scale ϵ by $W_\epsilon(V)$ and the actual solution by $\mathcal{G}_\epsilon(V)$. The augmented operator estimates in Paper I give the inverse on the orthogonal complement of the stationary coordinates and retain the stationary finite-dimensional factor explicitly. For the regular index-zero solution the complete realized operator is invertible.

The determinant comparison is constructed using the same finite-dimensional stabilization on the limiting and preglued problems. The patched kernel map identifies the two stabilized kernels for small ϵ . Cancelling the stabilization factors gives

$$(7.2) \quad \det D_V \longrightarrow \det L_{W_\epsilon(V)}.$$

At a constant reference configuration this is the positive identity. Coherence under gauge and strip gluing propagates the normalization to each component of the mixed configuration space, by the shifting argument of Proposition 5.3.

It remains to compare the preglued operator with the operator at the actual solution. A potential difficulty is that the chart used for the contraction estimate need not be the chart used to define the determinant bundle. The second derivative of an arbitrary fixed mixed chart need not have a uniform bound in the collapse norm. We use the following argument instead.

Lemma 7.1 (Transport around a square). *Let $D_{s,t}$ be a continuous family of Fredholm operators on a square. Suppose the operators on three edges are invertible. Parallel transport of orientations on the fourth edge carries the canonical orientation at its first endpoint to the canonical orientation at its second endpoint.*

Proof. The orientation double cover of the determinant bundle is trivial over the square. Transport around its boundary is therefore the identity. On each of the three invertible edges the canonical orientation is a continuous section, so transport is positive. The remaining edge must have the same property at its endpoints. $\square$

Proposition 7.2. *The map (7.2), followed by transport from $W_\epsilon(V)$ to $\mathcal{G}_\epsilon(V)$, preserves the sign of the zero-dimensional solution.*

Proof. Use the contraction chart of Paper I on one horizontal edge of a square. The straight coordinate segment from the preglue to the solution stays in the region where the derivative differs from the invertible approximate operator by a sufficiently small bounded operator. Consequently this edge is invertible. At the solution, change from the contraction realization to the admissible orientation realization by Lemma 5.2; this vertical edge is invertible. At the approximate configuration use the same source and target identifications as those defining (7.2); the comparison gives the other invertible vertical edge.

The family in the interior of the square consists of admissible Fredholm realizations. This uses Hypothesis 2.1 and the exact matching chart, including the conformal gauge row. The fourth edge is the path of orientation operators at the configurations between the preglue and the solution. We need not assert that this entire edge is invertible. Lemma 7.1 nevertheless transports the canonical orientation between its endpoints. Combining this with (7.2) proves the claim. $\square$

If the conformal metric in the collapse estimate is changed within the unperturbed collar class, the zero set of the mixed equation is unchanged. At a solution Lemma 5.2 compares the gauge rows by invertible paths, so the sign is unchanged. This permits use of the metric of Paper I without introducing a new orientation convention.

The construction is compatible with compact parameter families because its stabilizations and operator bounds may be chosen locally uniformly. Associativity of ordinary gluing identifies the determinant maps at broken ends. This proves Theorem B.

7.3. Vertex identifications and two-handles. In the triangle degeneration a completing vertex is identified with a product of a gauge path and a constant mixed pair. Normalize this identification using the corresponding end frames, with the outgoing coordinate translated in the positive direction on every arm. The constant pair has sign +1 and the gauge path uses the fixed object line. The resulting vertex identification has sign +1 at the reference configuration. Shifting and gluing show that it has that sign throughout its configuration component.

For the standard surgery triple let θ_G denote the distinguished gauge cycle in the auxiliary connected sum of $S^1 \times S^2$'s. Normalize its symplectic counterpart by

$$\theta_S = N_{12}\theta_G.$$

The unit diagonal and the usual filtration of that auxiliary complex identify this with the distinguished top cycle, with the chosen sign. The mixed triangle family has one endpoint equal to a gauge triangle followed by the mixed map, and the other endpoint is a symplectic triangle with the two mixed inputs. Proposition 7.2 compares their orientations. The broken boundary terms give a homotopy, hence

$$(7.3) \quad N_{02}\mu_G \simeq \pm\mu_S(N_{01} \otimes_R N_{12}).$$

Evaluating the second input on θ_G gives

$$(7.4) \quad N_{p_1}I_J^\#(W_2) \simeq \pm S_{W_2}N_{p_0}.$$

Changing bases conjugates this identity by the right-placed conversions and uses (3.2). Twisting the bundle on the two-handle changes the gauge triangle class but not the orientation argument; its relative bundle is retained throughout the mixed family.

For simultaneous gluing at two triangle ends, use one stabilization off both ends. The determinant comparison is independent of the order of gluing, up to the graded reordering already present in (3.4). Thus the triangle systems are coherent at corners, which is needed when the same product is later folded into a compression cup.

8. TORUS EXCISION OVER THE INTEGERS

8.1. The closed cut and its orientation data. The folded triangle has a closed gauge cut

$$Z \cong (-M) \# M^+ \# T^3.$$

The odd tori $T_a, T_b \subset Z$ are homologous, and the lifting group at this cut is generated by $s_Z = \text{PD}[T_a] \pmod{2}$. Excision along these tori gives

$$Z' = (-M)^\# \sqcup (M^+)^\#.$$

Write $E : C_G(Z) \rightarrow C_G(Z')$ for the gauge excision map and $\bar{E}$ for its reverse. The tensor complex at Z' uses the right convention.

The almost-complex structure induced by the folded triangle at Z need not equal the one induced by the mixed torus domain. A conversion at this interior cut is therefore required. We choose the structures so that

$$(8.1) \quad c_1(J)[T_a] \equiv 0 \pmod{4}.$$

For two such structures, the character (4.1) on s_Z is trivial. Consequently the conversion at Z is defined on the quotient complex.

Here is why the choice is available. On an oriented four-manifold with boundary, spin^c structures form an affine space for H^2 , and their first Chern classes change by twice that class. If the image of $H^2(B; \mathbb{Z}) \rightarrow H^2(T_a; \mathbb{Z})$ is $m\mathbb{Z}$, write $[T_a] = m\sigma$ modulo torsion. When m is odd, twisting changes half the Chern evaluation modulo two and permits (8.1). When m is even, the evaluation on σ is even: the restriction of $w_2(B)$ to its oriented three-dimensional boundary vanishes, and σ may be taken there. Thus the evaluation on T_a is already divisible by four. The almost-complex structure is then made cylindrical near Z . This argument is applied with the fixed torus germ away from the interior cut.

Convention 8.1. The determinant line of the conversion at an interior cut is placed after the complete input generator line and before the complete configuration determinant line. Its position is fixed relative to the entire configuration, independently of where the geometric cut is drawn.

Suppose the configuration is written as a sequence of glued pieces. Moving an odd conversion line with the cut through that sequence would move it past the determinant line of the intervening piece. The sign would then depend on its index. Convention 8.1 avoids this dependence: moving the line to a fixed reference position costs a sign determined by the entire configuration, which is unchanged by moving the cut. At index zero the alternative fixed placement after the complete configuration line agrees with this one.

8.2. Lifting groups and gluing multiplicities. It is useful to perform the gauge excision calculation before passing to a quotient. Let $\mathcal{G}_0(X)$ be the subgroup of bundle automorphisms with trivial lifting class. If a finite group ϕ_X acts freely and orientation preservingly on the relevant zero-dimensional solution set, its quotient count is

$$(8.2) \quad I(X)^{\phi_X}(\bar{a}_0, \bar{a}_1) = \frac{1}{|\phi_X|} \sum_{a_i \in \bar{a}_i} \#\mathcal{M}(X; a_0, a_1).$$

The numerator is divisible by $|\phi_X|$: it is a sum of free orbits, each with a constant sign.

Lemma 8.2. *Let X be connected and let Y be a closed hypersurface with cut manifold X_Y . Suppose the critical limits on Y have trivial stabilizer in the adjoint determinant-one gauge group. For each matched configuration on X_Y that glues, the resulting $\mathcal{G}_0(X)$ -classes form a torsor for*

$$K_Y = \ker(H^1(X; \mathbb{Z}/2) \longrightarrow H^1(X_Y; \mathbb{Z}/2)).$$

In particular their multiplicity is $|K_Y|$.

Proof. Choose representatives whose two limits agree under the fixed gluing identification. If two choices glue to the same determinant-one class on the cut manifold, their difference is an automorphism whose lifting class restricts to zero on X_Y . Conversely every such class changes the gluing while leaving the cut class fixed. The stabilizer condition makes this action free. Its transitivity follows by comparing the two gluing identifications on the two copies of Y . The Mayer–Vietoris sequence identifies K_Y with the image of the connecting

map from $H^0(Y; \mathbb{Z}/2)$. The analytic assertion that every required matched solution glues and that there are no others is Hypothesis 2.3. $\square$

A finite lifting group acts freely on the solutions in the present calculation. Indeed, a fixed solution would give a parallel automorphism. Its restriction to an irreducible end must be the identity once its lifting class there vanishes. A parallel automorphism which is the identity on an end is the identity everywhere on a connected cobordism. The almost-complex orientation construction preserves signs under the whole group. Thus (8.2) applies.

8.3. The capped comparison. Let $V = \mathcal{E} \cup_{Z'} \bar{\mathcal{E}}$ be the composite of excision and reverse excision. The standard local model is T^2 times the doubled excision polygon. It contains a nonseparating hypersurface $\Sigma = T^2 \times \gamma$. Cut along Σ to obtain V° , and cap the two new boundary components by $T^2 \times D^2$. The capped cobordism V' is a product on Z ; this is the geometric comparison in torus excision [6, Theorem 5.6].

The determinant-one complex on Σ has two critical points g_1, g_2 . The class $a = \text{PD}_\Sigma[T^2]$ interchanges them. They must be kept separate when the caps are attached: the caps carry the trivial lifting group, since a does not extend across their meridional disks.

Write $s = \text{PD}_Z[T_a] \pmod{2}$. In $H^1(V^\circ; \mathbb{Z}/2)$ let p be its pullback and h_1, h_2 the meridional classes of the two removed cap neighborhoods. The restrictions are

	Z_{in}	Z_{out}	Σ_{in}	Σ_{out}
p	s	s	0	0
h_1	s	0	a	0
h_2	s	0	0	a
$h_1 + h_2$	0	0	a	a

The group $H^\circ = \langle p, h_1, h_2 \rangle$ has order 8. It contains both groups needed on the cut cobordism, but is not itself the pullback of either of the groups on V and V' .

More precisely, ϕ_V has order 8, its image on V° is $\langle p, h_1 + h_2 \rangle$ of order 4, and its kernel is K_Σ of order 2. The capped product has group $\phi_{V'} = \langle p \rangle$ of order 2. To check these assertions, the two sides of the excision polygon give classes P_+, P_- and the disconnected intermediate cut gives a class c . Their restrictions are $p, p, h_1 + h_2$, respectively. They are independent on V , as seen by loops crossing each of the two polygon sides and the intermediate cut. Their sum $P_+ + P_-$ is dual to Σ , generates the restriction kernel, and is nonzero because Σ is nonseparating.

Choose invariant regularizing data on V° . Let

$$D(\alpha, \sigma; \beta, \tau) = \#\mathcal{M}(V^\circ; (\alpha, \sigma), (\beta, \tau)),$$

where $\sigma, \tau \in \{g_1, g_2\}$. Put $\Delta(\alpha, \beta) = D(\alpha, g_1; \beta, g_1)$. The action of $h_1 + h_2$ makes the two diagonal blocks equal; the action of h_2 identifies the off-diagonal block with $\Delta(s\alpha, \beta)$. Thus, after summing over endpoint lifts, diagonal and off-diagonal blocks have the same sum.

Each cap has one regular flat solution in its specified bundle and therefore invariant $\pm g_1$ (or the corresponding dual generator). The odd bundle on T^2 has no infinitesimal flat deformations, which gives regularity. The index-energy formula excludes a nonflat solution of index zero. Hence the cap coefficient is a unit over $\mathbb{Z}$.

Proposition 8.3. *Under Hypothesis 2.3, the quotient counts for V and V' have the same absolute multiplicity. In each fixed degree they agree up to the determinant-gluing sign.*

Proof. For V , self-gluing has multiplicity 2 and there are two equal diagonal blocks. Quotienting by the group of order 8 gives

$$\frac{2 \cdot 2}{8} \sum_{\alpha \in \bar{a}, \beta \in \bar{b}} \Delta(\alpha, \beta).$$

For V' , the caps have unit coefficients, their separating gluings have multiplicity 1, and the group has order 2. The result is

$$\frac{1}{2} \sum_{\alpha \in \bar{a}, \beta \in \bar{b}} \Delta(\alpha, \beta).$$

If the second cap selects g_2 , the symmetry of the off-diagonal block gives the same sum. The orientation signs are those in the gluing hypothesis. All group actions preserve the orientations, so no cancellation occurs inside an orbit. The two numerical factors agree. $\square$

In particular no division by two is performed on a Floer homology group. The fractions above express integer orbit counts. Computing the caps only in the rank-one quotient would lose the distinction between matching two lifts and matching two orbits.

The one-dimensional gluing family identifies these counts with the composite excision map. The same argument for the other composite gives the reverse comparison. Consequently $H(E)$ and $H(\bar{E})$ are isomorphisms. On a homogeneous class their composite is that class up to a sign depending on its degree. This is the only inverse-excision statement needed for the handle state below. By Lemma 3.1, the closed excision maps are homotopy equivalences (after shifting degrees if necessary). We do not replace a possibly degree-dependent sign by a single scalar on an ungraded complex.

8.4. The mixed excision identity. Let N_Z be the mixed map on the closed-cut configuration, and N_-, N_+ the vertex maps on the two separated ends. The symplectic output of N_Z is a Floer complex in a product target. Folding identifies its generators with pairs (x, y) in the reflected source and target. The identification of their orientation lines is denoted by $\mathcal{K}$; it is fixed by comparison at split constant reference configurations. If an input conversion has parity t_{in} , write $\mathcal{K}_R = \mathcal{K}\varepsilon^{t_{\text{in}}}$ for its closed-map convention.

Theorem 8.4 (Mixed excision, conditional). *Under the analytic hypotheses, with conversions at the closed cut placed as in Convention 8.1, one has*

$$(8.3) \quad \mathcal{K}_R N_Z \bar{E} \simeq \pm (N_- \otimes_R N_+).$$

For a homogeneous cycle $z \in C_G(Z)$ this implies

$$(8.4) \quad [\mathcal{K}_R N_Z z] = \pm [(N_- \otimes_R N_+) E z].$$

All terms are expressed at the fixed end bases.

Proof. Use the closed-cut gluing from Paper I to express $N_Z \bar{E}$ as the joined mixed count. The ordinary neck homotopy gives a signed chain homotopy by Lemma 3.3 and (5.3). Next vary the width and complete the torus ends. This is exactly the comparison asserted in Hypothesis 2.2. At the completed limit the zero-dimensional solutions are pairs of solutions on the two separate domains.

At split reference configurations the determinant identification is the product identification by Hypothesis 2.2(b). Coherence at gauge and strip ends and the shifting argument propagate this comparison to every component. Thus the only generator-dependent factor is the fixed folding identification $\mathcal{K}$ itself; there is no additional uncontrolled diagonal sign. Listing the

two output lines in order gives $N_- \otimes_R N_+$. The comparison families supply the homotopy in (8.3).

Apply (8.3) to Ez . The capped comparison says $[\bar{E}Ez] = \pm[z]$ for this homogeneous class. Applying N_Z gives (8.4). $\square$

Equality (8.4) is equality of homology classes of tensors. If the tensors differ by a boundary, applying the chain isomorphism (3.6) produces a boundary in the Hom complex. This is the mechanism that will turn the handle-state calculation into a chain homotopy. No inference from equality of induced maps on homology is involved.

9. THE COMPRESSION CUP BEFORE REFLECTION

9.1. The tensor supplied by the folded triangle. Let C be the compression cobordism for k one-handles, let p be the source presentation, and put $q = C \circ p$. The folded target is a product of a negatively oriented source moduli space and the target moduli space. The graph of restriction over C is a Lagrangian correspondence in this product. Two auxiliary height Lagrangians supply distinguished intersection cycles q_0^*, q_1^* .

The folded triangle product, with the conversion at Z inserted, gives a cycle

$$(9.1) \quad z_C = \mathcal{K}_R \mu_X^\Theta(q_0^* \otimes_R q_1^*) \in CF(\bar{p}) \otimes_R CF(q).$$

The triangle and the folding are defined by holomorphic curves and their transported orientation lines. Define

$$(9.2) \quad \check{U}_C = B_{CF(\bar{p})}(z_C) : CF(\bar{p})^\vee \longrightarrow CF(q).$$

For $k = 0$ denote this map by $B(p, r)$, where r is the upper height presentation. These maps have not yet used a symplectic reflection pairing.

The top cycles are normalized using the integral mixed maps on the auxiliary vertices: $q_i^* = \pm N_i \theta_i$. The sign is one sign of each distinguished cycle. The auxiliary complexes have the standard filtration, and the unit-diagonal calculation in Proposition 6.3 makes this an integral normalization, not merely a mod-two one.

9.2. The gauge handle state. Let B_k be the gauge filling whose closed boundary is Z , obtained from the folded triangle and its two auxiliary fillings. Its relative invariant is a homogeneous cycle $\Theta_{B_k} \in C_G(Z)$. Choose a single almost-complex structure on this filling and use its restrictions on the pieces. Convert at Z to the structure used for mixed torus excision. After excision the resulting state is a cycle on $C_G(-M) \otimes_R C_G(M^+)$.

Bending the source end of the one-handle cobordism W_k gives the identity

$$(9.3) \quad (\Phi_{G,M} \otimes_R 1)[E\Theta_{B_k}] = \pm[\text{ten}(I_J^\#(W_k))].$$

To check the sign convention, start with the product cylinder: its bent state is the coevaluation tensor. Glue W_k to its second end and use the ordered determinant gluing. This gives the tensor of the right-placed map. An odd endpoint conversion is already part of that map through (4.3); moving it to the other side would change the tensor by a grading operator. The residual signs in (9.3) are therefore scalar. The same argument holds for $k = 0$, when the cobordism map is gauge continuation.

9.3. The cup formula. Define the transposed reflected mixed map

$$(9.4) \quad n_p = \text{ev}_{C_G(M)}^{-1} (N_{\bar{p}} \Phi_{G,M}^{-1})^\vee : CF(\bar{p})^\vee \longrightarrow C_G(M).$$

It is an even chain isomorphism, using the paired choices of gauge base. This definition uses gauge duality and the mixed map at the reflected presentation; it makes no assumption about symplectic duality.

Proposition 9.1 (Cup formula). *Under the analytic hypotheses,*

$$(9.5) \quad \check{U}_C \simeq \pm N_q I_J^\#(W_k) n_p.$$

In particular, for a lower datum p and a compatible upper height datum r ,

$$(9.6) \quad B(p, r) \simeq \pm N_r \text{cont}_G(p, r) n_p.$$

Proof. Apply the integral triangle comparison to the two distinguished input cycles. It identifies the folded triangle output, up to a boundary and one sign, with $N_Z \Theta_{B_k}$, using the conversion at Z on both sides. Apply (8.4) to the homogeneous cycle Θ_{B_k} . The result is

$$[z_C] = \pm [(N_{\bar{p}} \otimes_R N_q) \mathbf{E} \Theta_{B_k}].$$

Equation (9.3) identifies this with the tensor of $N_q I_J^\#(W_k) n_p$. The identification follows directly from (3.6) and the graded double-dual map in (9.4). Since the difference of the tensors is a boundary, their images under $B_{CF(\bar{p})}$ differ by δK . This proves (9.5). The same calculation with $k = 0$ gives (9.6). $\square$

This proves a formula for the cup tensor without selecting the reflection pairing needed for the source of Paper I's map. The next sections construct that pairing and show that it gives the ordinary symplectic continuation when $k = 0$.

10. SPHERE CHARACTERS AND VARIATION OF PRIMARY DATA

10.1. The character of an orientation system. For a transverse pair $P = (P_0, P_1)$ let $\mathcal{P}(P_0, P_1)$ be the path space of paths in $\mathbf{M}_h$ starting on P_0 and ending on P_1 . For $h \geq 2$, the compression-body Lagrangians are simply connected and have vanishing second homotopy group. The relative homotopy classes of strip configurations with fixed ends are consequently a torsor for $\pi_2(\mathbf{M}_h) \cong \mathbb{Z}$ in the lift convention of [1, Section 2]. Fix its positive generator S_h ; attaching it changes the index by four. For $h = 1$ the moduli space is a point and the sphere-character assertions below are empty.

The determinant line of a sphere operator, together with the inverse determinant of the matching tangent space at its node, has its complex orientation. Compare node gluing with the coherent strip orientation Γ^P . The ratio is a sign

$$(10.1) \quad \chi^P : \pi_2(\mathbf{M}_h) \longrightarrow \{1, -1\}.$$

Associativity of node gluing makes it a character. Multiplying generator lines by signs does not change it: the two endpoint factors occur in both sides of the comparison. Multiplying them by their grading signs has the same property. Thus the character is information about the configuration-level coherent system, beyond the signs of its differential.

At a transverse point x , a second description is available. Attach a fixed representative sphere to the constant mixed pair C_x and compare its transported mixed orientation with the product of the canonical orientation of C_x and the complex node orientation. Denote the ratio by $\chi(h, x; d)$, where h is the primary datum and d contains the regularizing and almost-complex data.

Lemma 10.1. *At regular data, $\chi(h, x; d) = \chi^P(S_h)$ for every transverse intersection point x . The ratio is constant along any compact smooth family of primary data carrying a persistent transverse intersection point. It is also independent of small regularizing data at fixed primary data and fixed gauge base.*

Proof. Glue a strip to the constant pair and attach the sphere away from the ordinary gluing collar. The two possible orders of these operations induce the same determinant-line isomorphism. The sphere and node factors are complex, hence even as real graded lines, so commuting them with the strip factor gives no sign. Cancelling the strip orientation gives equality of the pointed and strip definitions. A strip configuration exists between any two intersection points, in some relative homotopy class; it need not solve the holomorphic equation. Applying the same argument to that configuration shows independence of x .

For a family with a persistent point, the constant-pair operators, the transported orientation, and the node-gluing map vary continuously by Hypothesis 2.1. The ratio is a continuous function to $\{1, -1\}$ and is therefore constant. To vary regularizing data, use the segment to zero data. The perturbations are supported off the strip end, and the constant-pair operators stay invertible. Transporting their canonical orientations along the segment proves the last assertion. Compatible almost-complex structures can be varied through the same argument, with the gauge base held fixed. $\square$

The persistent point need only remain transverse at that point. Other intersections of the two Lagrangians may become degenerate. The lemma is about determinant lines over configurations and therefore does not require a regular moduli space along the family.

10.2. A common ball of primary perturbations. Let $\mathcal{C}_i$ be the finite real span of the sign-sensitive holonomy functions on C_i^o , modulo constants. Use the norm

$$(10.2) \quad \begin{aligned} \|h_i\|_i &= \sup_B \|\nabla_B h_i\|_{L^\infty} + \sup_{B \neq B'} \frac{\|\nabla_B h_i - \nabla_{B'} h_i\|_{L^2}}{\|B - B'\|_{L^2}}, \\ \|h\| &= \max(\|h_0\|_0, \|h_1\|_1). \end{aligned}$$

The suprema are taken over smooth connections in a fixed determinant bundle. Cylinder functions have finite norm by the holonomy estimates of [1, Proposition 6.1]. If the norm vanishes, the gradient vanishes on the affine connection space, so the function is constant; thus this is a norm modulo constants.

Choose a finite trace family F_0 whose restrictions to the compact flat-connection moduli spaces give embedding coordinates. In the model $L(C_i) \cong \mathrm{SU}(2)^{h-1}$, quaternion coordinates can be obtained from sign-sensitive traces using the fixed holonomies on the odd torus. Such traces distinguish central sign changes; passing to unsigned $\mathrm{SO}(3)$ traces would lose that property. For a finite family F containing F_0 , put

$$\mathcal{U}(F) = \{h \in \mathrm{span} F : \|h\| < r_*\}, \quad \mathfrak{B} = \bigcup_F \mathcal{U}(F).$$

The radius is independent of the additional functions in F , for the fixed splitting and metric data. These sets are nested under inclusion of finite families and are convex.

Proposition 10.2. *The radius $r_* > 0$ can be chosen so that every h in this ball has the following properties.*

- (i) *The perturbed flat moduli space of each compression body is a smooth compact manifold identified with its unperturbed moduli space; its boundary restriction is an embedded Lagrangian close to the unperturbed one.*

- (ii) The connection representatives and their first derivatives in the moduli parameters are close in L_1^2 and L^2 , respectively, uniformly as $\|h\| \rightarrow 0$.
- (iii) The functions from F_0 remain embedding coordinates on these perturbed moduli spaces.
- (iv) If the two boundary Lagrangians intersect transversely, the corresponding closed gauge critical points are nondegenerate and are exactly the lifts of those intersections.

Proof. The norm gives $\|F_B\|_{L^\infty} \leq \|h_i\|_i$ for an h_i -flat connection. If a sequence with norm tending to zero escaped every gauge neighborhood of the flat moduli space, Uhlenbeck compactness on the compact compression body would give a subsequence weakly converging in $W^{1,4}$, after gauge, to a flat connection. The $W^{1,4}$ embedding in C^0 is compact in three dimensions. A Coulomb–Neumann slice relative to the flat limit and the elliptic estimate for (d_B^*, d_B) then improve this to convergence in L_1^2 , a contradiction. The boundary slice and compactness statements are the versions for a manifold with boundary in [13].

In a slice at a flat connection split into the finite-dimensional harmonic space and its complement. The linearized perturbed-flat operator on the complement has a bounded right inverse. The second term of (10.2) makes the holonomy part a small Lipschitz perturbation in L^2 , so the contraction gives a graph over the harmonic space. Differentiating its fixed-point equation in the finite-dimensional parameter gives the asserted first-derivative control. Hodge decomposition and boundary elliptic regularity identify these solutions with the smooth perturbed-flat solutions; the exact regularity inputs are recorded in Appendix A. Compactness of the flat moduli space makes the thresholds uniform over finitely many slice charts.

For a trace function τ , its bounded Lipschitz gradient extends the identity

$$\tau(B') - \tau(B) = \int_0^1 \langle \nabla_{B+s(B'-B)} \tau, B' - B \rangle ds$$

to L^2 connections. Differentiating on the graph family shows that the trace-coordinate map remains C^1 -close to its unperturbed embedding. Openness of embeddings of a compact manifold proves (iii). The matching exact sequence for the two compression-body boundary problems identifies the kernel of the closed Hessian with the tangent intersection of the Lagrangians; hence transversality gives (iv), as in [1, Section 2.3]. $\square$

No radius controlling all higher derivatives of every cylinder function is asserted here. The norm controls the two estimates actually used in this proposition. On a fixed finite-dimensional perturbation family all higher smooth seminorms are finite and are handled separately in compact-family regularity statements.

10.3. Paths with only birth–death singularities. Enlarge F to include the monomials of degree at most five in the trace coordinates F_0 . These remain cylinder functions: a monomial is a smooth invariant function of the joint finite holonomy family.

Lemma 10.3. *Two regular primary data in $\mathfrak{B}$ can be joined within a common $\mathcal{U}(F)$ by a smooth path with finitely many singular parameters. At each singular parameter there is one quadratic tangency, with versal normal form*

$$(10.3) \quad S_t(x, y) = x^3 + \sigma(t - t_*)x + Q(y), \quad \sigma \in \{1, -1\},$$

where Q is nondegenerate. All other intersections are transverse.

Proof. Start with the segment from h_a to h_b and perturb it to

$$h_t = (1 - t)h_a + th_b + \beta(t)c,$$

where β vanishes near the endpoints and is positive on the interior where singularities may occur. The segment is compact in the open convex ball, so small c keeps the path in the ball.

Variation of the holonomy function induces an exact Lagrangian deformation; its Hamiltonian on the perturbed Lagrangian is the restriction of that function to the connection moduli space. The trace coordinates remain embedding coordinates by Proposition 10.2. Polynomials of degree at most three in embedding coordinates realize arbitrary third jets at one point. Polynomials of degree at most five realize arbitrary second jets at two distinct points. For the latter assertion, choose an affine function ℓ with $\ell(a) = 0$ and $\ell(b) = 1$. Given a second jet at b , multiply ℓ^3 by the quadratic Taylor polynomial at b of that jet divided by ℓ^3 . The product has degree at most five, realizes the prescribed jet at b , and vanishes to order three at a . Use $(1 - \ell)^3$ in the same way for the jet at a , then add the two polynomials. Restriction of ambient jets to an embedded submanifold is surjective.

In a Weinstein chart, varying one Lagrangian changes the relative generating function by this Hamiltonian to first order. The map on its jets is triangular with identity diagonal. Parametric transversality therefore excludes higher singularities, two simultaneous tangencies, and tangencies with zero crossing speed: these strata have codimension at least two. A generic one-parameter path meets only the codimension-one birth–death stratum, transversely. Its parameter-dependent splitting normal form is (10.3).

The singular-parameter set is closed by compactness of the Lagrangians. Each singular parameter is isolated by the normal form, and the endpoint neighborhoods are regular. It is consequently finite. No uniform separation of the two points in the two-point jet argument is required; transversality is applied on the open configuration space of distinct pairs. $\square$

Lemma 10.4. *Suppose $H(C_G(M)) \neq 0$. The path in Lemma 10.3 has a finite subdivision with regular subdivision points such that each subinterval carries a continuous transverse intersection point throughout the subinterval.*

Proof. Every regular primary datum has at least one intersection. Otherwise its gauge complex would have no generators by Proposition 10.2(iv), contradicting invariance and nonvanishing of the gauge homology.

At a birth–death time the local pair in (10.3) is absent on one side. If it were the only intersection at the singular time, compactness would force the nearby regular intersection set on that side to be empty. Thus another intersection exists at that time and is transverse. The implicit function theorem continues it across the wall.

On a closed interval with no walls, the incidence set of intersections is a compact one-manifold projecting locally diffeomorphically to the interval. It is a finite covering, hence a disjoint union of sections. Extend the chosen transverse point from each wall to neighboring regular subdivision times. This gives the required finite collection of persistent sections. $\square$

Theorem 10.5 (Constancy of the sphere character). *Fix a splitting, metric data and gauge base, and suppose $H(C_G(M)) \neq 0$. Then $\chi^P(S_h)$ is independent of the regular datum P with primary data in $\mathfrak{B}$.*

Proof. First remove the regularizing data at both endpoints using Lemma 10.1. Join the primary data by the subdivided path of Lemma 10.4. On each subinterval the pointed character is constant along its persistent transverse point. At a regular subdivision time the pointed character equals the global character for any choice of point, so one may switch to the persistent point for the next subinterval. Chaining these equalities gives constancy along the whole path. Restore the endpoint regularizing data. The same argument allows variation of the compatible symplectic almost-complex structure at a fixed gauge base. $\square$

The argument uses nonvanishing only to find a point persisting through each birth–death. It does not assume that individual generators persist along an arbitrary path, or that the regular locus of the perturbation ball is connected.

11. REFLECTION OF THE TRANSPORTED ORIENTATIONS

11.1. Reflection and sphere gluing. The reflected presentation $\bar{p}$ interchanges the two compression bodies. A strip is reflected by

$$\bar{u}(s, t) = u(-s, 1 - t).$$

This map preserves the orientation of the strip domain and keeps the surface almost-complex structure. It reverses the translation direction. The target symplectic manifold retains the original orientation; the reversed order of the Lagrangians supplies the dual complex. Mixed solutions themselves are not identified by this formula. The orientation at $\bar{p}$ is transported through its own mixed domain.

For a strip from x to y , the reflection of its coherent orientation has differential sign

$$(11.1) \quad \text{sign}_{\text{dual}}(\bar{u}) = -(-1)^{|y|} \text{sign}(u).$$

The minus sign is the reversed translation vector; the other sign is the graded transpose. Thus this reflected system is the one for which the paired generator bases give the graded dual differential.

Attaching S_h changes the index by four. The determinant factor at a sphere node is complex and even. The additional transpose sign is $(-1)^{4|y|} = 1$. Consequently reflection of a coherent strip system preserves its sphere character. What remains is to compare this reflected system with the system transported through the reflected mixed domain.

11.2. One folded comparison. Fix one height-fold instance with lower primary datum $\tilde{p}$ and upper height datum r , both in the admissible ball. Choose the height after the radius of that ball, as in Section 13. Double the folded triangle and attach its auxiliary caps. The resulting disk has an incoming strip end modeled on $\tilde{p}$ and an outgoing end modeled on r . Its transported orientation is coherent with the reflected incoming strip system and the outgoing strip system.

Proposition 11.1. *Under the analytic hypotheses,*

$$\chi^{\tilde{p}}(S_h) = \chi^r(S_h).$$

Proof. Choose a smooth reference configuration c_0 on the doubled disk. Attach a representative sphere on the incoming strip end, then move its node through the disk to the outgoing end. The two resulting configurations lie in the same relative class: the path of the node and its trivialization explicitly gives a homotopy. If a product target is used in the folded description, perform this operation on the appropriate factor before unfolding. The other factor is kept fixed.

There is a continuous orientation on the determinant bundle along this path. Its construction on the folded domain uses the collapse transport of Section 7, the closed-cut conversion, and Hypothesis 2.2. At the two ends, coherence identifies it with the strip-node gluing orientations. No Koszul sign is introduced by moving the complex node factor. Their ratios are therefore equal, giving the assertion.

This comparison uses arbitrary smooth configurations, not only holomorphic solutions. To justify coherence where the fold is attached to prescribed strip configurations, select one reference configuration for each of the finitely many generator pairs and their required sphere

classes. Choose the height and neck lengths so that ordinary gluing applies to this finite set. The determinant-line construction then extends over each configuration component by path independence. The regularity and completion hypotheses are needed on all the domains in this finite construction; a comparison only at the constant solution would not suffice. $\square$

Changing the almost-complex base on the closed cut changes the output generator lines by a fixed diagonal conversion. That conversion cancels in a sphere ratio. Thus the character comparison is unchanged by the conversion required in Section 8, although that conversion remains essential for the cup identity as a chain-level formula.

Corollary 11.2 (Reflection invariance). *For every regular datum p in the admissible ball at an object with nonzero integral gauge homology,*

$$\chi^{\bar{p}}(S_h) = \chi^p(S_h).$$

Proof. Apply Theorem 10.5 on $-M$ to compare $\bar{p}$ with $\tilde{p}$. Apply Proposition 11.1 to compare the latter with r . Apply character constancy on M to compare r with p . The reflected ball is the same ball under the chosen reflection identification. Gauge duality makes nonvanishing for M and $-M$ equivalent. $\square$

11.3. The reflection pairing. Compare on the strips of $\bar{p}$ its own transported system and the dual system of (11.1). Their ratio is constant on smooth configuration components and multiplicative under concatenation. Choose one reference intersection point. A configuration from it to any other intersection determines a sign $\kappa_p(x)$. Changing that configuration changes its class by a sphere; Corollary 11.2 says that the ratio on this sphere is one. Hence the ratio on a strip from x to y is $\kappa_p(x)\kappa_p(y)$, independent of the strip class.

Proposition 11.3. *The map*

$$(11.2) \quad \Phi_p : CF(\bar{p}) \longrightarrow CF(p)^\vee, \quad o_x^{\bar{p}} \longmapsto \kappa_p(x)(o_x^p)^\vee$$

is an even chain isomorphism. The configuration-level comparison determines it up to one sign. The pairings may be chosen so that

$$(11.3) \quad \Phi_{\bar{p}} = \Phi_p^\vee \text{ ev}.$$

Proof. Equation (11.1) and the ratio just computed give exactly the dual differential under (11.2). The paired gauge bases make the sum of the two endpoint parities zero, so the map is even. Any two comparisons have constant ratio: configurations exist between every pair of intersection points, and the ratio on all those configurations fixes the relative generator signs. Only the initial reference sign is free.

Reflect twice on a strip of index n with endpoints x, y . The two transpose signs multiply to $(-1)^{n(|x|+|y|)} = (-1)^n$. The corresponding change of the paired basis is the sign in the graded evaluation map. Thus the pairing obtained by the second reflection is exactly $\Phi_p^\vee \text{ ev}$, after the common reference sign is fixed. This proves (11.3) at configuration level. $\square$

A chain-map condition alone would not determine κ_p up to one sign. If the differential splits into two subcomplexes, one can change the sign on one summand and still obtain a chain map. The preceding construction uses all strip configurations, including those which are not solutions, to exclude that extra freedom.

Put

$$(11.4) \quad \widehat{\Phi}_p = \Phi_p^\vee \text{ ev} : CF(p) \longrightarrow CF(\bar{p})^\vee.$$

This is the reflection pairing which will be composed with the cup map (9.2).

12. CONTINUATION AND THE DIAGONAL COMPRESSION CUP

12.1. Coherent orientations with prescribed ends. A continuation datum from p to q has a strip-like surface with Lagrangian boundary conditions varying on a compact interval and equal to those of p, q at its ends. The almost-complex and Hamiltonian data are likewise constant outside that interval. For generators x, y , let $\mathcal{B}_{p,q}(x, y)$ be its smooth configuration space, with the prescribed exponential limits. A coherent orientation system assigns relative orientations to the determinant lines over these spaces and commutes with attaching strips at either end.

The space of configurations in a fixed relative homotopy class is path connected. One may first choose a continuous relative homotopy, smooth it relative to the fixed ends and labels, and then make it constant beyond common end cutoffs. Local charts are given by an exponential map for a metric making the boundary labels totally geodesic. The topology in Section 2 makes these paths continuous. The relative classes are a torsor for the same sphere group as fixed-label strips, and the index changes by four under its generator. In particular there is at most one index-zero component for fixed endpoints.

Lemma 12.1 (Rigidity of coherent systems). *Two coherent continuation orientation systems with the same end systems differ by one sign. Such a system exists if and only if the sphere characters at the ends agree.*

Proof. The ratio of two systems is constant on each configuration component. If a strip is attached at either end, the common end orientation cancels, so the ratio does not change. Choose one reference configuration from x_0 to y_0 . For any x, y , choose strips from x to x_0 and from y_0 to y . Attaching these strips transports the ratio to a configuration with ends x, y . Any difference in relative class can be supplied by a sphere class on either strip. Since the end systems are the same, this also leaves the ratio unchanged. Thus there is only one sign for all configurations.

For existence choose a reference orientation and extend it by the same gluing procedure. The determinant bundle on each configuration component is orientable: the boundary Lagrangians are the parallelizable compression- body Lagrangians, and the remaining sphere factors have complex orientation. The only compatibility left when the gluing paths are changed is moving a sphere class from the incoming end to the outgoing end. The two extensions agree precisely when the two sphere characters agree. This also proves necessity by moving the node of one sphere through a continuation configuration. The construction is continuous because the determinant line is a line bundle over the smooth configuration space. $\square$

In an index-zero comparison a finite form of the lemma suffices. Choose one strip for each input generator and one for each output generator, in the classes needed to reach the index-zero component. Coherence on that finite collection fixes the ratios of all zero-dimensional counts. This form permits the height and neck thresholds to be chosen after a finite set of configurations, without a claim of uniform estimates over the whole configuration space.

Theorem 10.5 supplies existence for p, q in the same admissible ball. For a lower datum and an auxiliary upper height datum, the folded configuration itself supplies a coherent system and hence the required character equality. The latter observation applies even when the auxiliary upper datum is outside the character-constancy ball, provided it satisfies the separate primary regularity threshold.

12.2. Parity and continuation maps. For a continuation configuration c from x to y , the number

$$d(p, q) = |y| - |x| - \text{ind } D_c \pmod{2}$$

is independent of the configuration and its end generators. It is additive under composition of continuation data. The configuration description by sphere classes shows its independence of the chosen datum. For the paired gauge bases of this paper, $d(p, q) = 0$ at the data used in the comparison. Here one can use the integral cup formula to check the parity without assuming that integral nonvanishing implies mod-two nonvanishing.

Indeed, at a fold pair (p, r) the map $B(p, r)$ is homogeneous and (9.6) identifies it up to homotopy with the even isomorphism $N_r \text{cont}_G(p, r)n_p$. On nonzero integral homology this map cannot be both even and odd. Thus its degree is zero. The doubled configuration has the same index as the continuation problem, and the gauge reflection pairing is even; hence $d(p, r) = 0$. A common upper height datum and additivity give $d(p, q) = 0$. All uses of the integral cup formula in this argument precede the diagonal-cup identification and therefore do not use the continuation properties being proved.

With a coherent system define

$$\text{cont}_S(p, q) : CF(p) \longrightarrow CF(q)$$

by the signed zero-dimensional continuation count. It is even and commutes with the differential. The compactification of a one-dimensional continuation moduli space proves this exactly as in Proposition 6.1.

Proposition 12.2. *The continuation maps satisfy, up to sign and chain homotopy,*

$$(12.1) \quad \text{cont}_S(q, r) \text{cont}_S(p, q) \simeq \pm \text{cont}_S(p, r),$$

$$(12.2) \quad \text{cont}_S(p, p) \simeq \pm 1.$$

Their homotopy classes are independent of the continuation datum. They are homotopy equivalences.

Proof. For a long strip neck, ordinary gluing identifies the zero-dimensional solutions with pairs of continuation solutions. The determinant gluing is coherent, so the signed matrix is the composite matrix. The glued orientation system has the same end systems as the chosen direct continuation system. Lemma 12.1 compares them by one sign. Varying the neck length gives the chain homotopy in (12.1).

A homotopy of continuation data, constant near the ends, gives the corresponding homotopy by its oriented one-dimensional boundary. The pure symplectic compactness and gluing used here are included in Hypothesis 2.1(d). For constant data the zero-dimensional solutions are constant strips, with canonical positive orientation. This normalizes the unit. Apply composition to a datum and its reverse to obtain a homotopy inverse. $\square$

12.3. Reflection of continuation. There is no translation quotient for a continuation configuration. Reflecting it therefore has the graded transpose sign but no additional translation sign. Using the pairings of Proposition 11.3 gives

$$(12.3) \quad \text{cont}_S(\bar{p}, \bar{q}) \simeq \pm \Phi_q^{-1} \text{cont}_S(q, p)^\vee \Phi_p.$$

To verify coherence of the reflected system, glue an input strip, a continuation configuration, and an output strip, and reflect the three pieces in reverse order. If their indices are n_u, n_c, n_w , the reordering sign is $(-1)^{n_u n_c + n_u n_w + n_c n_w}$. Combining it with the three transpose signs leaves $(-1)^{n_u d(p, q)} = 1$. The generator corrections κ_p and κ_q identify the reflected end systems with

the own transported systems. Rigidity then gives the one sign in (12.3). This argument also shows why parity zero is needed here.

12.4. Identification of the diagonal cup. For $k = 0$ the compression correspondence is the diagonal. Double the folded triangle, glue the two distinguished caps, and unfold along that diagonal. The resulting surface is a continuation strip. This is the geometric identification used in Paper I. To obtain an integral statement we compare its coherent systems before counting.

Proposition 12.3 (Diagonal cup). *At every admissible fold pair (p, r) ,*

$$(12.4) \quad B(p, r) \hat{\Phi}_p \simeq \pm \text{cont}_S(p, r).$$

Proof. The doubled triangle has the transported fold orientation. Attach the two auxiliary cap configurations, each normalized by its unit relative invariant. Unfolding identifies the resulting real Cauchy–Riemann operator with the continuation operator, with its reflected input paired through $\hat{\Phi}_p$. The normal direction at the diagonal and the ordered input lines give precisely the graded transpose convention used in (11.4).

On the finite configuration set from Lemma 12.1, ordinary strip gluing and the orientation comparisons of the collapse and folded domains show coherence with the prescribed input and output systems. Attach input and output strips successively; this reduces product-strip coherence to the one-factor comparisons. After extending over the configuration components, the pulled-back system is a continuation system. Rigidity compares it with the chosen system by one sign.

For sufficiently long cap necks, the gluing theorem gives a bijection of zero-dimensional solutions and preserves these orientations. A cap of index zero has one regular solution with unit coefficient, so no extra multiplicity occurs. The counts are therefore the same up to that sign. Varying the cap length and the triangle data to the ones defining $B(p, r)$ gives a chain homotopy through the one-dimensional moduli spaces. The preglue-to-solution path is within the pure section chart, and determinant transport on that path preserves the canonical orientation of a regular solution. This is the pure symplectic gluing input in Hypothesis 2.1. $\square$

This completes Theorem C. The statement uses the compression cup of the manuscript. It does not replace continuation by an inverse of a chosen mixed map.

12.5. Objects with vanishing homology. Suppose $H(C_G(M)) = 0$. Lemma 3.1 and Proposition 6.3 make both $C_G(M, p)$ and $CF(M, p)$ contractible for every presentation. Every closed map from or to a contractible complex is a boundary: compose it with a contracting homotopy and use (3.2). Hence there is exactly one morphism between it and any other object in the homotopy category.

At such an object we take the continuation and reflection maps needed for the comparison to be zero representatives of those unique morphisms. In particular a reflection map here is a homotopy equivalence between zero objects, not an invertible matrix. We assert neither character constancy nor existence of the same configuration-level reflection pairing at this object. Every identity involving it is interpreted in the homotopy category.

This convention also handles a word passing through an acyclic intermediate object: its class is zero, as is the corresponding gauge composite. Reduction modulo two preserves a contracting homotopy. Thus Paper I’s maps incident to that object are null-homotopic as well.

Nonzero mod-two gauge homology implies nonzero integral gauge homology, since an acyclic finite free integral complex would be contractible. The standard mod-two framed calculation therefore places rational homology spheres and $\#^k(S^1 \times S^2)$ in the nonvanishing case [11, Corollary 1.4 and Section 7.8]. We do not use the converse implication: a nonzero finite odd-torsion integral group need not be detected modulo two.

13. CHOICES, PRIMARY NATURALITY AND HANDLE MAPS

13.1. Order of the auxiliary choices. All choices in a fixed comparison involve finitely many presentations. The following order avoids requiring a perturbation to be simultaneously small relative to a parameter chosen after it.

- (1) Fix the splittings, metric data, paired gauge bases and trace-coordinate families for all objects and their reflections. Choose a common primary regularity threshold r_H for this finite set, as in Proposition 10.2.
- (2) Choose the first auxiliary heights t_*, t'_* so their height perturbations and small generic corrections have norm less than r_H . These data are used as upper auxiliary presentations.
- (3) Choose the character-constancy radius r_* below r_H and inside the small-primary neighborhoods required for those heights. Use the same choice for a presentation and its reflection.
- (4) Choose smaller heights t'' for the direct duality argument, t_f for the reflection fold, and t_C for the compression cup. Their height data lie inside the corresponding balls. Choose the lower datum of a cup after t_C , sufficiently small relative to that height.
- (5) Select the finite reference configurations needed for coherence. Choose their necks and the collapse scales, then the regularizing data, below all thresholds for those fixed geometric data. In particular, an auxiliary upper datum is a full presentation with its own vertex-map thresholds, not merely a pair of transverse Lagrangians.
- (6) Choose the weights inside all end gaps in that instance, below the minimum of their decay rates. The completed-domain estimates are required at the same choices by Hypothesis 2.1.

Every choice has a positive threshold once the preceding finite data have been fixed. Auxiliary upper data may lie outside the final character ball; the folded comparison supplies their coherent continuation system directly. The lower data and the handle target where reflection is used lie inside the relevant ball.

13.2. Small-primary naturality. We now prove the comparison for changes of primary data. All maps in the next two propositions are even.

Proposition 13.1. *For regular p, q in one admissible ball,*

$$(13.1) \quad N_q \operatorname{cont}_G(p, q) \simeq \pm \operatorname{cont}_S(p, q) N_p.$$

Proof. Choose one upper height presentation a of $-M$ generic for $\bar{p}$ and $\bar{q}$. Apply the cup formula and Proposition 12.3 to both fold pairs. They give

$$\begin{aligned} \operatorname{cont}_S(\bar{p}, a) &\simeq \pm N_a \operatorname{cont}_G(\bar{p}, a) n_{\bar{p}} \widehat{\Phi}_{\bar{p}}, \\ \operatorname{cont}_S(\bar{q}, a) &\simeq \pm N_a \operatorname{cont}_G(\bar{q}, a) n_{\bar{q}} \widehat{\Phi}_{\bar{q}}. \end{aligned}$$

All factors being eliminated are homotopy equivalences. Cancel the common upper datum using continuation composition. Thus

$$\begin{aligned} \text{cont}_S(\bar{p}, \bar{q}) &\simeq \pm (n_{\bar{q}} \widehat{\Phi}_{\bar{q}})^{-1} \text{cont}_G(\bar{p}, \bar{q}) \\ &\quad \cdot n_{\bar{p}} \widehat{\Phi}_{\bar{p}}. \end{aligned}$$

Use the reflection formula (12.3) on the symplectic side and its gauge analogue from (4.5). The symmetric conventions give

$$\widehat{\Phi}_{\bar{p}} = \Phi_p, \quad \Phi_{G, Mn_{\bar{p}}} = N_p^\vee,$$

and the same equalities at q . Cancelling these pairings transforms the displayed identity into

$$\text{cont}_S(q, p)^\vee \simeq \pm (N_p \text{cont}_G(q, p) N_q^{-1})^\vee.$$

Transpose and rename p, q to obtain (13.1). Every double-dual map occurs together with its inverse. Because the maps here are even, the cancellations introduce no grading-dependent sign. $\square$

This proof uses the same pairing in the diagonal-cup and reflection identities. Choosing an unrelated chain-level duality isomorphism would leave a diagonal automorphism after the cancellation.

13.3. Mixed duality. Put $N_p^\dagger = n_p \widehat{\Phi}_p$. This is the transpose of the reflected mixed map, with the gauge and symplectic pairings inserted.

Proposition 13.2. *For every regular p in the admissible ball,*

$$(13.2) \quad N_p N_p^\dagger \simeq \pm 1.$$

Proof. First take p small enough for a fold at the height t'' , with upper datum r in the same ball. The two cup identities give

$$\text{cont}_S(p, r) \simeq \pm N_r \text{cont}_G(p, r) n_p \widehat{\Phi}_p.$$

By primary naturality the right side is homotopic, up to sign, to $\text{cont}_S(p, r) N_p N_p^\dagger$. Cancel the continuation equivalence.

For a general datum q in the ball, transpose primary naturality for the reflected presentations. The symmetric pairings give

$$N_q^\dagger \text{cont}_S(p, q) \simeq \pm \text{cont}_G(p, q) N_p^\dagger.$$

Combine this with primary naturality and (13.2) at p :

$$N_q N_q^\dagger \text{cont}_S(p, q) \simeq \pm \text{cont}_S(p, q).$$

Cancel the continuation once more. The contractible case has the interpretation of Section 12.5. $\square$

13.4. One- and three-handle squares. At the compatible handle data define the integral compression cup by

$$(13.3) \quad U_C = \check{U}_C \widehat{\Phi}_p : CF(p) \longrightarrow CF(q), \quad q = C \circ p.$$

Define the cap by reflected transposition. If $\bar{C}$ is the reflected compression and $\check{U}_{\bar{C}} : CF(p)^\vee \rightarrow CF(\bar{q})$ is its unpaired cup, put

$$(13.4) \quad \check{V}_C = \text{ev}_{CF(p)}^{-1}(\check{U}_{\bar{C}})^\vee, \quad V_C = \check{V}_C \widehat{\Phi}_q.$$

Then $V_C : CF(q) \rightarrow CF(p)$. Equivalently it is the transpose of the paired reflected cup with the two pairings inserted. These are the orientations of the compression cup and cap counts of Paper I. At other presentations they are conjugated by the geometric continuation maps.

Theorem 13.3. *For the one-handle cobordism W_k and its corresponding three-handle cobordism W_3 , one has*

$$(13.5) \quad U_C N_p \simeq \pm N_q I_J^\#(W_k),$$

$$(13.6) \quad V_C N_q \simeq \pm N_p I_J^\#(W_3).$$

The identities hold at all admissible endpoint presentations.

Proof. The cup formula and mixed duality give

$$U_C N_p \simeq \pm N_q I_J^\#(W_k) n_p \widehat{\Phi}_p N_p \simeq \pm N_q I_J^\#(W_k),$$

which proves (13.5). For the cap, transpose the cup formula at the reflected compression. The graded evaluation maps and $\Phi_{G,-M} = \Phi_{G,M}^\vee \text{ev}$ give the gauge transpose of the reversed handle map. Mixed duality at q cancels the reflected mixed map. In detail, in the homotopy category,

$$\begin{aligned} V_C N_q &= \check{V}_C \widehat{\Phi}_q N_q \\ &\simeq \pm \check{V}_C n_q^{-1} \\ &\simeq \pm N_p \Phi_{G,-M}^{-1} I_J^\#(\overline{W}_3)^\vee \Phi_{G,-M^+} \\ &\simeq \pm N_p I_J^\#(W_3). \end{aligned}$$

The only potentially odd map in the transpose comparison is the handle map; all pairings and mixed maps are even. Its degree is retained in the graded transpose, so any remaining sign is scalar.

Finally conjugate both handle maps by continuations from arbitrary admissible endpoints to the compatible handle data. Primary naturality moves the mixed maps past these continuations, and gauge functoriality identifies the gauge composite with the same cobordism. Homotopies compose by (3.2). $\square$

14. DESCENT TO COBORDISMS AND PROOF OF THE MAIN THEOREM

14.1. The remaining changes of presentation. The geometric presentation moves are the ones in Paper I. We record how the integral orientations enter them. A change of regularizing data uses the family defining Proposition 6.3. A change of strip-end almost-complex structure uses the analogous mixed continuation family; the end systems are identified by the determinant transport in Proposition 5.3. For a change of gauge base, both the gauge and symplectic generator lines are converted by the same right-placed maps. The reference constant pair has no extra line, so its orientation comparison commutes with this conversion.

An isomorphism of decorated data pulls back the equation, its linearization and the shifting construction. Choose pulled-back stabilizations in the determinant construction. The kernel gluing maps then commute with the isomorphism, and the mixed square commutes exactly on chains at the pulled-back data. A subsequent continuation gives the square at the originally chosen data, up to sign and homotopy. This applies to frame changes and diffeomorphisms preserving the torus decoration.

A Heegaard stabilization is realized by the cancelling one- and two-handle pair of Paper I. Compose (13.5) and (7.4). The decorated gauge cobordism is the identity, so its map is

continuation up to sign and homotopy. The reverse stabilization is treated by the cancelling two- and three-handle pair. Passing through a common stabilization treats changes of metric data and isotopies of the splitting. The order of choices in Section 13 allows the new vertices to satisfy the integral height and ball conditions. No orientation conclusion is inferred from cancellation of unsigned counts alone.

Thus every elementary presentation arrow $e : p \rightarrow q$ has a symplectic map S_e constructed from the curves of Paper I and a square

$$(14.1) \quad N_q G_e \simeq \pm S_e N_p.$$

The gauge arrows G_e represent the relevant decorated cobordism or identity continuation. This includes both even presentation moves and possibly odd handle arrows.

14.2. Projective descent.

Lemma 14.1. *Suppose a connected presentation category maps to a cobordism category and every cobordism has an elementary word. Let G_e be a projective gauge functor on these words. Suppose N_p are isomorphisms in a homotopy category and the elementary maps satisfy (14.1). Then S_d depends only on the represented cobordism and the endpoint presentations, up to sign and homotopy.*

Proof. For a word $d = e_m \cdots e_1$, compose the squares to obtain

$$S_d N_{p_0} \simeq \pm N_{p_m} G_{e_m} \cdots G_{e_1}.$$

Every composition of homotopies is valid by (3.2). If d' represents the same cobordism, gauge functoriality identifies the two gauge words up to sign and homotopy. Cancel N_{p_0} to identify S_d and $S_{d'}$.

For words representing the identity cobordism this also gives the transitive system of comparison maps between presentations. The identity word is the identity map, and concatenation of words gives composition up to sign. Hence these systems descend to a projective functor and the classes of N_p give a projective natural isomorphism. $\square$

Proof of Theorem A. Proposition 6.3 gives the integral mixed isomorphisms. Equation (7.4), Theorem 13.3, and the presentation comparisons above supply all elementary squares. Generation and connectivity are the geometric presentation results of Paper I, with the additional smallness conditions arranged as in Section 13. Proposition 4.4 and Lemma 14.1 therefore give (1.1) and the projective functoriality.

The mixed maps preserve the chosen object gradings. On homology, the comparison with the homogeneous gauge map therefore gives the asserted parity of the symplectic map. No separate assertion of homogeneity for every chosen chain representative of a long word is needed.

Modulo two all determinant orientations give the same counts. The conversions become the identity on the generator basis, the signs in the tensor and dual formulas disappear, and the pairing (11.2) becomes the paired-basis reflection pairing. Thus the geometric maps reduce to those of Paper I at the same data. At an acyclic object the zero representatives reduce to maps homotopic to those of Paper I by Section 12.5. $\square$

14.3. What remains for an unconditional extension. Theorem A is a conditional integral comparison. Removing its analytic hypotheses requires a proof on the actual completed mixed domains of uniform regularity, uniform slice and weight estimates, and continuity of

the orientation identifications in the torus completion. The nonseparating self-gluing statement must also be verified with the lifting groups and ordered determinant lines used here. These steps are separate from the orbit-count algebra in Proposition 8.3.

Several further extensions ask different questions. Exact composition signs require more than projectivizing the maps; the regrading discussed after (4.6) indicates the extra grading information involved. An identification with integral homology-oriented instanton maps requires a comparison of orientation systems on the framed quotient. An identification with integral quilted Floer maps requires a comparison with relative-spin orientations on the compression correspondences. None of these identifications follows from the conditional theorem by reducing modulo two.

APPENDIX A. ANALYTIC INPUTS AND THEIR PRECISE USES

This appendix records the analytic statements used in the draft and separates the ordinary inputs from the completed-domain hypotheses. All references to Paper I mean its version of 25 September 2026. The present paper does not supply an independent proof of that manuscript's unsigned collapse, torus excision, or generation results.

A.1. Mixed Fredholm theory and shifting. The mixed linearization couples the anti-self-dual operator to a real Cauchy–Riemann operator by the linearized matching condition. For fixed admissible ends it is Fredholm on the prescribed weighted spaces. The local regularity and compactness statements for the original mixed equation are in [2]; the vertex Fredholm and constant-pair statements used here are in [1, Theorems 4.12, 4.17 and Proposition 3.19]. The mixed shifting statements are [1, Lemmas 4.62 and 4.64]. Their use on the completed and folded domains, including the relative family form in the smooth topology, is explicitly covered by Hypothesis 2.1.

At an ordinary strip end the asymptotic operator is $J\partial_t$ with the totally real endpoint conditions. At a gauge end it is the extended Hessian. A weight δ must be separated from the relevant spectrum; at a decaying end it must be smaller than the decay rate of the configuration and of the linearized kernel and cokernel elements. At a mixed end the stationary modes are split off before the weight is imposed. This last finite factor remains in determinant gluing and may not be discarded.

A.2. Boundary Hodge theory and holonomy regularity. For a flat orthogonal bundle on a compact compression body, the input needed in Proposition 10.2 is the Hodge decomposition on $W^{1,2}$ with normal boundary condition. Its harmonic fields are smooth and finite dimensional, and its exact part is the covariant differential of $W^{2,2}$ sections. The Dirichlet Green operator for $d_B^*d_B$ preserves smoothness, so $1 - d_B G_D d_B^*$ is a smoothness-preserving projection onto coclosed forms. These statements follow from the flat covariant de Rham case of [8, Section 1.6 and Theorems 3.10–3.11]; the normal and tangential versions are interchanged by Hodge star.

The boundary elliptic estimate has the form

$$\|b\|_{W^{1,p}} \leq C(\|d_B b\|_{L^p} + \|d_B^* b\|_{L^p} + \|b\|_{L^p})$$

for $*b|_{\partial C} = 0$. For smooth fixed B , adjoint bundle coefficients add only bounded zeroth-order terms in local trivializations. The compactness and Coulomb-slice inputs are Theorems A, D(a) and F of the introductory statements in [13].

Finite holonomy functions have bounded gradients and Lipschitz gradient differences as used in (10.2) [1, Proposition 6.1]. Their higher derivatives obey the local tame estimates obtained by differentiating parallel transport. Combined with the boundary elliptic estimate,

these give the bootstrap from the slice fixed point to the smooth perturbed-flat solution. For holonomy families with moving basepoints, restriction and composition are performed in the smooth topology, not asserted bounded on arbitrary low-order Sobolev spaces.

A.3. Pure symplectic gluing. The continuation and cap arguments require the following form of gluing. For a finite collection of regular configurations with nondegenerate strip ends, there is a sufficiently long neck such that every nearby zero-dimensional solution is the unique correction of a preglued configuration. The correction tends to zero with the neck length, and the linearized gluing map identifies the determinant orientations. The estimates are uniform over compact families of the fixed regular configurations. The strip construction is provided by the gluing analysis of [12]; for quilted surfaces see [9, 10]. Exponential decay is used at every ordinary end; in the transverse case the full-strip statement is [12, Proposition 2.3.1].

Compactness in dimensions zero and one requires the monotonicity and index restrictions for the actual Lagrangians and correspondences. The compactified one-dimensional space then has the endpoint and ordinary broken-strip strata used for continuation. The monotone quilted compactness and family constructions are in [15, Theorems 3.9 and 4.1] and [10, Section 3]. Their applicability to every pure symplectic domain used here is included in Hypothesis 2.1(d). A cohomological composition identity alone is not used as a substitute for this moduli-space gluing statement.

A.4. The unresolved uniform comparisons. The following are the forms of the remaining estimates needed to remove the hypotheses. For the completed gauge slices one needs a uniform Poincare estimate

$$\|\zeta\|_{L^2} \leq C \|d_A \zeta\|_{L^2}$$

on the specified gauge subspace, including the end framing conditions. Together with the higher elliptic estimates it must give a positive Coulomb slice radius independent of the growing torus length. For mixed linearizations one needs uniform regularity on the joined, separate and closed-cut domains, with the same asymptotic weights and with the finite-dimensional stationary coordinates retained. Finally the preglued-to-solution paths must lie in the configuration charts where the determinant transport is defined. Pointwise regularity at each finite length proves none of these uniform assertions by itself.

For the gauge self-gluing, one must additionally check that the compactified one-dimensional stretching family has no omitted end, and that the kernel of the restriction of lifting classes is exactly the gluing fiber in Lemma 8.2. The finite group calculation in Section 8 then gives integral unit multiplicity. Keeping these analytic statements separate makes clear what is assumed by the first draft and what the orientation algebra proves from them.

APPENDIX B. A SIGN CALCULATION FOR PROJECTIVE COMPOSITION

The conversion of a strictly commuting map F of degree f to a closed map is $F^r = F\varepsilon^f$. For another strictly commuting map G of degree g , the relation $\varepsilon^f G = (-1)^{fg} G\varepsilon^f$ gives

$$F^r G^r = (-1)^{fg} F G \varepsilon^{f+g} = (-1)^{fg} (FG)^r.$$

Thus this convention change is compatible with composition after projectivization. The factor is scalar, despite the grading operator used in converting each map.

For a strict homotopy $F - F' = dH + Hd$, where $|H| = f + 1$, the correctly converted homotopy is $H\varepsilon^f$. Indeed

$$\delta(H\varepsilon^f) = dH\varepsilon^f - (-1)^{f+1} H\varepsilon^f d = (dH + Hd)\varepsilon^f.$$

Using $H\varepsilon^{f+1}$ instead would give the wrong sign. This calculation is useful whenever a gauge composition homotopy is translated into the Hom complex convention.

Lastly, if $F_i - G_i = \delta K_i$ and the maps are closed, a homotopy between two composites is obtained by replacing one factor at a time and using (3.2). For example, for three factors one may take

$$K_3 F_2 F_1 + (-1)^{|G_3|} G_3 K_2 F_1 + (-1)^{|G_3|+|G_2|} G_3 G_2 K_1.$$

Applying δ telescopes to $F_3 F_2 F_1 - G_3 G_2 G_1$. This is the chain-level algebra used in Lemma 14.1.

REFERENCES

- [1] A. Daemi, K. Fukaya and M. Lipyanskiy, *Lagrangians, $SO(3)$ -instantons and the Atiyah–Floer conjecture*, arXiv:2109.07032.
- [2] A. Daemi, K. Fukaya and M. Lipyanskiy, *Lagrangians, $SO(3)$ -instantons and mixed equation*, Geom. Funct. Anal. **34** (2024), 659–732; arXiv:2109.07038.
- [3] S. K. Donaldson, *The orientation of Yang–Mills moduli spaces and 4-manifold topology*, J. Differential Geom. **26** (1987), 397–428.
- [4] S. K. Donaldson, *Floer Homology Groups in Yang–Mills Theory*, Cambridge Tracts in Math. **147**, Cambridge Univ. Press, 2002.
- [5] A. Floer and H. Hofer, *Coherent orientations for periodic orbit problems in symplectic geometry*, Math. Z. **212** (1993), 13–38.
- [6] P. B. Kronheimer and T. S. Mrowka, *Khovanov homology is an unknot-detector*, Publ. Math. Inst. Hautes Études Sci. **113** (2011), 97–208.
- [7] P. B. Kronheimer and T. S. Mrowka, *Knots, sutures, and excision*, J. Differential Geom. **84** (2010), 301–364.
- [8] R. Kupferman and R. Leder, *Elliptic pre-complexes, Hodge-like decompositions and overdetermined boundary-value problems*, arXiv:2304.08977v3.
- [9] S. Ma’u, *Gluing pseudoholomorphic quilted disks*, arXiv:0909.3339.
- [10] S. Ma’u, K. Wehrheim and C. T. Woodward, *A_∞ functors for Lagrangian correspondences*, Selecta Math. (N.S.) **24** (2018), 1913–2002; arXiv:1601.04919v5.
- [11] C. W. Scaduto, *Instantons and odd Khovanov homology*, J. Topol. **8** (2015), 744–810.
- [12] T. Simčević, *A Hardy space approach to Lagrangian Floer gluing*, Ph.D. thesis, ETH Zürich, Diss. ETH No. 22118, 2014; arXiv:1410.5998.
- [13] K. Wehrheim, *Uhlenbeck Compactness*, EMS Series of Lectures in Mathematics, European Mathematical Society, Zürich, 2004.
- [14] K. Wehrheim and C. T. Woodward, *Orientations for pseudoholomorphic quilts*, arXiv:1503.07803.
- [15] K. Wehrheim and C. T. Woodward, *Pseudoholomorphic quilts*, J. Symplectic Geom. **13** (2015), 849–904.
- [16] K. Wehrheim and C. T. Woodward, *Quilted Floer cohomology*, Geom. Topol. **14** (2010), 833–902.
- [17] B. J. Wuebben, *Cobordism maps and Atiyah–Floer isomorphisms for framed instanton homology*, manuscript, 25 September 2026, <https://github.com/bwuebben/donaldson-atiyah-floer/tree/main/af-cob-1>.
- [18] A. Zinger, *The determinant line bundle for Fredholm operators: construction, properties, and classification*, Math. Scand. **118** (2016), 203–268.